Change in tobacco use among 13–15 year olds between 1999 and 2008: findings from the Global Youth Tobacco Survey

Charles W. Warren[1], Veronica Lea[1], Juliette Lee[1], Nathan R. Jones[2], Samira Asma[1] and Matthew McKenna[1]

Abstract: Tobacco use is the leading preventable cause of death and disease in the world; yet little is known about the levels or patterns of youth tobacco use on a global basis. The purpose of this paper is to focus on change in youth tobacco use using data from 100 sites that have conducted repeat Global Youth Tobacco Surveys (GYTS). The GYTS is a school-based survey that collects data from students aged 13–15 years using a standardized methodology for constructing the sample frame, selecting schools and classes, and processing data. GYTS is conducted in school classes using self-administered anonymous data collection. The GYTS sample produces representative, independent, cross-sectional estimates for each sampling frame. Of the 100 sites surveyed, 61 reported no change over time in prevalence of cigarette smoking, likewise in 50 of the 97 sites with data on use of other tobacco products there was no change. However, 34 sites reported an increase in other tobacco use. This appears to be attributed to waterpipe, an emerging trend in tobacco use. Evidence was found supporting the idea that tobacco use among adolescent girls is likely increasing. The global tobacco control effort continues to face many challenges in reversing the tobacco epidemic. Few countries have implemented comprehensive tobacco control legislation laid out by the World Health Organization. The few countries that have adopted some of these proven policies can serve as examples in achieving positive results in curbing the tobacco epidemic. (Global Health Promotion, 2009; Supp (2): pp. 38-90)

Introduction

Tobacco use is the leading preventable cause of death and disease in the world (1); total tobacco-attributable deaths are projected to increase from approximately 5M per year today to over 8M per year by 2030 (2). Two global initiatives were initiated in the late 1990s to address the tobacco epidemic. First, the World Health Organization (WHO) initiated the WHO Framework Convention on Tobacco Control (WHO FCTC) in 1997 (3). The WHO FCTC is the world's first public health treaty on tobacco control and has been ratified by over 160 WHO Member States. The WHO FCTC 'provides the principles and context for policy development, planning of interventions and mobilization of political and financial resources for tobacco control' (2). By ratifying the Treaty countries make a commitment to join the fight against the tobacco epidemic.

Second, in 1998 in response to countries' concerns about the lack of tobacco use data for youth, WHO, the Centers for Disease Control and Prevention, and the Canadian Public Health Association developed and initiated the Global Youth Tobacco Survey

1. US Centers for Disease Control and Prevention (CDC), Atlanta, Georgia. Correspondence to Charles W. Warren. (wcw1@cdc.gov)
2. University of Wisconsin.
For the GYTS Country Research Coordinators; and GYTS support at WHO Headquarters and Regional Offices, Canadian Public Health Association, and CDC (Appendix A).

Global Health Promotion 1757-9759; Supp (2): 38–90; 342192 Copyright © The Author(s) 2009, Reprints and permissions: http://www.sagepub.co.uk/journalspermissions.nav DOI: 10.1177/1757975909342192 http://ghp.sagepub.com

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

(GYTS) (4). The GYTS provides systematic global youth tobacco use data which can be used by countries to monitor tobacco use and meet the provisions of Article 20 of the WHO FCTC.

> Article 20: Research, surveillance and exchange of information.
>
> 2. The Parties shall establish, as appropriate, programmes for national, regional and global surveillance of the magnitude, patterns, determinants and consequences of tobacco consumption and exposure to tobacco smoke. Towards this end, the Parties should integrate tobacco surveillance programmes into national, regional and global health surveillance programmes so that data are comparable and can be analyzed at the regional and international levels, as appropriate. (3)

To date, 154 WHO Member States and 13 other populations have initiated the GYTS; 100 have conducted second rounds, and 9 have conducted third rounds.

The purpose of this report is to focus on changes in youth tobacco use. GYTS data from 100 sites that have conducted repeat surveys can be used to determine what, if any, changes in tobacco use have occurred among youth (95 WHO Member States; one territory [US Virgin Islands], two geographic regions [Gaza Strip and West Bank], one special administrative region [Macau], and one commonwealth [(Northern Mariana Islands]). For each site, comparisons between Time 1 and Time 2 (or Time 3) surveys will be made to identify patterns of change. Two indicators of tobacco use will be compared:

- Prevalence of current cigarette smoking (at least one day during the past month)
- Prevalence of current use of other tobacco products [such as waterpipe, smokeless tobacco, chewing tobacco] (use during the past month)

Methodology

The GYTS is a school-based survey that collects data from students aged 13–15 years using a standardized methodology for constructing the sample frame, selecting schools and classes, and processing data (4). The GYTS uses a two-stage cluster sample design that produces weighted estimates of students in grades associated with students 13–15 years old. Each sampling frame includes all schools (usually public and private) in a geographically defined area containing any of the identified grades. At the first stage, the probability of schools being selected is proportional to the number of students enrolled in the specified grades. At the second sampling stage, classes within the selected schools are randomly selected. All students in selected classes attending school the day the survey is administered are eligible to participate. Student participation is voluntary and anonymous using self-administered data collection procedures. The GYTS sample design produces representative, independent, cross-sectional estimates from each sampling frame. In most cases, repeat GYTS follow the same sample design as Time 1; however, if the educational system changed, then adjustments to the sampling frame were made to maintain comparability over time.

Repeat GYTS have been conducted in countries across the six WHO Region Offices: African Region (AFR), Eastern Mediterranean Region (EMR), European Region (EUR), Region of the Americas (AMR), South-East Asia Region (SEAR), and Western Pacific Region (WPR). The GYTS data in this report include 20 member states in AFR (12 national and 8 sub-national) for a total of 20 sites; 15 member states in EMR (14 national and 1 sub-national) and 2 geographic regions for a total of 17 sites; 18 member states in EUR (15 national and 3 sub-national) for a total of 18 sites; 29 member states in AMR (16 national and 13 sub-national) and 1 territory for a total of 30 sites; 6 member states in SEAR (4 national and 2 sub-national) for a total of 6 sites; 7 member states in WPR (4 national and 3 sub-national) and 1 special administrative region and 1 commonwealth for a total of 9 sites (Table 1). In total, separate, representative GYTS were conducted in 95 WHO member states and 5 other areas. For the remainder of this article, discussion will focus on the 100 sites.

Variable Definitions

The core GYTS instrument includes 54 questions covering seven categories: tobacco use, knowledge and attitudes regarding tobacco, secondhand smoke (SHS) exposure, pro- and anti-tobacco media and advertising exposure, desire for cessation, access and availability to obtain tobacco, and having been taught in school about the harmful effects of tobacco use.

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 1. Number of sites included in this study by World Health Organization (WHO) region, WHO member state, geographic region, territory, special administrative region, or commonwealth

Number of Sites	*WHO Regions**						
	AFR	*EMR*	*EUR*	*AMR*	*SEAR*	*WPR*	*TOTAL*
WHO Member States	20	15	18	29	6	7	95
Geographic Region		2					2
Territory				1			1
Special Administrative Region						1	1
Commonwealth						1	1
TOTAL	20	17	18	30	6	9	100

*WHO Regions:
AFR – African Region
EMR – Eastern Mediterranean Region
EUR – European Region
AMR – Region of the Americas
SEAR – South-East Asia Region
WPR – Western Pacific Region

Research coordinators are encouraged to develop and add questions to the core questionnaire to gather information important to their country or WHO region. The final country questionnaires are translated into local languages as needed and back-translated to check for accuracy. GYTS country research coordinators conduct focus groups of students aged 13–15 years to further test the accuracy of the translation and student comprehension of the questions. The core GYTS questions used and indicators created in this report are consistent over time.

This report describes changes between 1999 and 2008 in two important tobacco use indicators: current cigarette smoking and use of other tobacco products. Current cigarette smokers are defined as students who answered one or more days to the question 'During the past 30 days (one month), on how many days did you smoke cigarettes?'. Current users of other tobacco products are defined as students who answered 'yes' to the question 'During the past 30 days (one month), have you ever used any form of tobacco products other than cigarettes (e.g. waterpipe, chewing tobacco, snuff, dip, cigars, cigarillos, little cigars, pipe)?'. Since 2006, the core GYTS questionnaire has included separate questions to distinguish between smoked and smokeless forms of other tobacco products. To maintain compatibility with earlier surveys, questions about tobacco products other than cigarettes were combined to form a general indicator of non-cigarette tobacco use for this report.

Data Analysis

A weighting factor is applied to each student record to adjust for non-response (by school, class, and student) and variation in the probability of selection at the school and class levels. A final adjustment sums the weights by grade and gender to the population of school children in the selected grades in each sample site. The GYTS sample design produces representative, independent, cross-sectional estimates for each site. SUDAAN, a software package for statistical analysis of complex survey data, was used to calculate weighted prevalence estimates and standard errors (SE) of the estimates (95% confidence intervals [CI]) were calculated from the SEs) (5). *T*-tests were used to determine differences between subpopulations (6, 7). In this paper, differences in proportions are considered statistically significant if the *t*-test *p* value was < 0.05.

Response Rates[i]

Among the 100 sites, 209 surveys were completed by the end of 2008: 100 initial surveys, 100 second round, and 9 third round (Table 2). Of the 209

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 2. Response rates by World Health Organization (WHO) region, WHO member state, geographic region, territory, special administrative region, or commonwealth–GYTS 1999–2008

			*Response Rates**					
Survey Year	*WHO Region and Site*	*National or Subnational Sample*	*Sch RR*	*Stu RR*	*Class RR*	*Overall RR*	*Percent of Girls in Survey Population*	*No. aged 13–15 years*
AFRICAN REGION								
2001	Botswana	National	96.0	95.6	100.0	91.7	55.6	1,084
2008	Botswana	National	100.0	96.0	100.0	96.0	56.1	1,611
2001	Burkina Faso	Ouagadougou	91.3	67.0	100.0	61.2	50.7	989
2006	Burkina Faso	Ouagadougou	100.0	94.7	100.0	94.7	50.4	1,356
2000	Ghana	National	100.0	83.1	100.0	83.1	44.9	1,088
2006	Ghana	National	96.7	88.5	100.0	85.6	45.4	5,185
2001	Kenya	National	100.0	97.1	100.0	97.1	49.3	3,093
2007	Kenya	National	100.0	89.4	100.0	89.4	51.7	6,768
2002	Lesotho	National	100.0	88.5	100.0	88.5	58.3	1,989
2008	Lesotho	National	88.0	94.6	100.0	83.2	57.0	1,604
2000	Malawi	Blantyre	92.3	85.2	100.0	78.6	51.3	783
2005	Malawi	Blantyre	100.0	85.6	100.0	85.6	51.5	1,151
2001	Mali	Bamako	100.0	86.3	100.0	86.3	43.1	1,079
2008	Mali	Bamako	100.0	86.0	100.0	86.0	47.6	1,032
2001	Mauritania	National	88.5	85.2	100.0	75.4	49.3	1,644
2006	Mauritania	National	94.3	88.3	100.0	83.3	46.0	2,040
2003	Mauritius	National	95.8	92.7	100.0	88.9	51.5	1,609
2008	Mauritius	National	96.0	82.5	100.0	79.2	52.9	1,286
2002	Mozambique	Maputo City	100.0	88.3	100.0	88.3	53.9	912
2007	Mozambique	Maputo City	92.0	84.1	100.0	77.4	51.7	1,381
2001	Niger	National	100.0	80.2	100.0	80.2	47.2	1,192
2006	Niger	National	100.0	84.0	100.0	84.0	49.0	1,055
2000	Nigeria	Cross River State	90.0	85.7	100.0	77.1	48.6	914
2008	Nigeria	Cross River State	100.0	96.0	100.0	96.0	50.0	349
2002	Senegal	National	100.0	92.3	100.0	92.3	42.3	2,157
2007	Senegal	National	92.0	89.3	100.0	82.1	50.4	1,611
2002	Seychelles	National	100.0	90.9	100.0	90.9	50.5	942
2007	Seychelles	National	100.0	86.0	100.0	86.0	50.7	844
1999	South Africa	National	76.9	85.5	100.0	65.7	52.6	2,579
2002	South Africa	National	92.3	68.1	100.0	62.9	52.9	4,325
2001	Swaziland	National	100.0	83.6	100.0	83.6	52.3	2,841
2005	Swaziland	National	100.0	86.5	99.8	86.3	53.6	7,628
2002	Togo	National	96.0	96.5	100.0	92.6	34.3	2,220
2007	Togo	National	100.0	91.5	100.0	91.5	38.0	1,947
2002	Uganda	Kampala	95.0	84.9	100.0	80.7	51.7	1,811
2007	Uganda	Kampala	96.0	85.9	100.0	82.4	50.3	1,337

* Response rates:
Sch RR–the school response rate is calculated as the number of participating schools divided by the number of selected schools
Stu RR–the student response rate is calculated as the number of participating students divided by the number of students enrolled in a class
Class RR–the class response rate is calculated as the number of participating classes divided by the number of classes selected
Overall RR–the overall response rate is calculated as the school rate times the class rate times the student rate
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 2. (Continued)

Survey Year	WHO Region and Site	National or Subnational Sample	Response Rates*				Percent of Girls in Survey Population	No. aged 13–15 years
			Sch RR	Stu RR	Class RR	Overall RR		
2002	Zambia	Lusaka	96.0	87.5	100.0	84.0	48.2	1,241
2007	Zambia	Lusaka	96.0	88.8	100.0	85.2	56.8	1,362
1999	Zimbabwe	Harare	100.0	83.0	100.0	83.0	52.4	621
2003	Zimbabwe	Harare	100.0	85.4	100.0	85.4	51.0	1,433
2008	Zimbabwe	Harare	88.0	92.8	100.0	81.7	51.4	1,181
EASTERN MEDITERRANEAN REGION								
2001	Egypt	National	100.0	96.3	100.0	96.3	47.3	2,215
2005	Egypt	National	90.2	85.4	100.0	77.0	43.6	2,898
2000	Gaza Strip§§	Gaza Strip	100.0	95.8	100.0	95.8	48.3	1,940
2005	Gaza Strip§§	Gaza Strip	100.0	94.5	100.0	94.5	50.3	1,395
2003	Iran	National	92.2	95.4	100.0	87.9	53.0	3,841
2007	Iran	National	92.3	93.1	100.0	85.9	44.0	1,153
1999	Jordan	National	91.0	92.2	100.0	83.9	49.9	2,847
2003	Jordan	National	100.0	89.1	100.0	89.1	48.8	3,683
2007	Jordan	National	100.0	91.6	100.0	91.6	50.9	1,550
2001	Kuwait	National	100.0	94.8	100.0	94.8	52.4	4,205
2005	Kuwait	National	100.0	88.7	100.0	88.7	53.9	2,636
2001	Lebanon	National	98.0	98.3	100.0	96.4	56.1	3,017
2005	Lebanon	National	98.0	99.2	100.0	97.2	52.6	2,431
2003	Libya	National	98.0	99.0	100.0	97.0	49.6	1,174
2007	Libya	National	100.0	94.1	100.0	94.1	49.9	1,243
2001	Morocco	National	98.0	94.8	100.0	92.9	43.2	2,008
2006	Morocco	National	98.0	93.5	100.0	91.6	44.7	1,991
2002	Oman	National	100.0	96.9	100.0	96.9	49.1	1,099
2007	Oman	National	100.0	96.8	100.0	96.8	46.3	1,214
2004	Qatar	National	92.0	91.9	100.0	84.5	54.1	2,229
2007	Qatar	National	96.0	90.9	100.0	87.3	63.1	943
2004	Somalia	Somaliland	88.0	94.2	100.0	82.9	36.7	374
2007	Somalia	Somaliland	96.0	90.2	100.0	86.6	31.6	897
2001	Sudan	National	94.0	94.2	100.0	88.5	50.8	1,734
2005	Sudan	National	92.0	93.2	100.0	85.7	51.7	2,831
2002	Syria	National	100.0	98.3	100.0	98.3	41.8	3,278
2007	Syria	National	100.0	83.7	100.0	83.7	48.2	1,621
2001	Tunisia	National	100.0	94.1	100.0	94.1	50.2	2,942
2007	Tunisia	National	100.0	92.4	100.0	92.4	49.6	1,499

* Response rates:
Sch RR–the school response rate is calculated as the number of participating schools divided by the number of selected schools
Stu RR–the student response rate is calculated as the number of participating students divided by the number of students enrolled in a class
Class RR–the class response rate is calculated as the number of participating classes divided by the number of classes selected
Overall RR–the overall response rate is calculated as the school rate times the class rate times the student rate
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 2. (Continued)

Survey Year	WHO Region and Site	National or Subnational Sample	Response Rates*				Percent of Girls in Survey Population	No. aged 13–15 years
			Sch RR	Stu RR	Class RR	Overall RR		
2002	United Arab Emirates	National	100.0	95.1	100.0	95.1	51.3	2,285
2005	United Arab Emirates	National	100.0	93.1	100.0	93.1	49.9	10,821
2000	West Bank§§	West Bank	98.7	94.8	100.0	93.5	52.2	4,387
2005	West Bank§§	West Bank	100.0	95.6	100.0	95.6	50.7	1,305
2003	Yemen	National	100.0	84.3	100.0	84.3	37.8	9,040
2008	Yemen	National	100.0	83.5	100.0	83.5	33.2	650
EUROPEAN REGION								
2003	Bosnia and Herzegovina	National	100.0	91.8	100.0	91.8	49.3	6,960
2008	Bosnia and Herzegovina	National	99.2	92.1	100.0	91.4	51.9	9,117
2002	Bulgaria	National	100.0	91.1	100.0	91.1	50.1	1,763
2008	Bulgaria	National	100.0	87.3	100.0	87.3	48.7	3,241
2003	Croatia	National	100.0	93.4	100.0	93.4	50.8	3,212
2007	Croatia	National	100.0	90.9	100.0	90.9	48.7	3,531
2002	Czech Republic	National	100.0	88.0	100.0	88.0	49.7	3,678
2007	Czech Republic	National	100.0	84.7	100.0	84.7	46.9	3,191
2003	Estonia	National	95.7	81.7	100.0	78.2	54.1	4,307
2007	Estonia	National	92.0	78.1	100.0	71.8	48.1	2,330
2003	Georgia	National	100.0	85.3	100.0	85.3	50.6	3,410
2008	Georgia	National	100.0	81.7	100.0	81.7	52.4	891
2003	Hungary	National	98.5	87.2	100.0	85.9	54.6	3,205
2008	Hungary	National	92.5	88.2	100.0	81.6	49.5	3,139
2004	Kyrgyzstan	National	100.0	98.8	100.0	98.8	58.2	3,434
2008	Kyrgyzstan	National	100.0	93.2	100.0	93.2	51.1	3,004
2002	Latvia	National	100.0	81.3	100.0	81.3	49.0	1,803
2007	Latvia	National	98.0	83.0	100.0	81.4	54.4	2,476
2001	Lithuania	National	100.0	82.4	100.0	82.4	49.5	1,752
2005	Lithuania	National	100.0	82.8	100.0	82.8	52.5	1,646
2004	Montenegro	National	100.0	87.0	100.0	87.0	55.4	1,966
2008	Montenegro	National	100.0	93.2	100.0	93.2	51.6	3,299
1999	Poland	Urban	96.5	87.4	100.0	84.4	54.5	1,297
2003	Poland	Urban	100.0	81.4	100.0	81.4	51.7	911
2004	Republic of Moldova	National	100.0	89.1	100.0	89.1	53.6	3,977

* Response rates:
Sch RR–the school response rate is calculated as the number of participating schools divided by the number of selected schools
Stu RR–the student response rate is calculated as the number of participating students divided by the number of students enrolled in a class
Class RR–the class response rate is calculated as the number of participating classes divided by the number of classes selected
Overall RR–the overall response rate is calculated as the school rate times the class rate times the student rate
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 2. (Continued)

Survey Year	WHO Region and Site	National or Subnational Sample	Response Rates* Sch RR	Stu RR	Class RR	Overall RR	Percent of Girls in Survey Population	No. aged 13–15 years
2008	Republic of Moldova	National	100.0	84.3	100.0	84.3	54.5	3,501
1999	Russian Federation	Moscow	99.0	86.0	100.0	85.2	50.9	3,157
2004	Russian Federation	Moscow	80.0	86.0	100.0	68.8	46.6	796
2003	Serbia	National	100.0	89.8	100.0	89.8	49.3	3,506
2008	Serbia	National	100.0	89.7	100.0	89.7	52.5	3,350
2003	Slovakia	National	98.3	87.4	100.0	85.9	50.6	3,639
2007	Slovakia	National	100.0	86.1	100.0	86.1	51.1	4,176
2003	Slovenia	National	95.0	89.1	100.0	84.6	55.8	3,817
2007	Slovenia	National	95.0	87.5	100.0	83.1	48.2	2,516
1999	Ukraine	Kiev	100.0	81.4	100.0	81.4	46.8	2,680
2005	Ukraine	Kiev	95.0	73.4	100.0	69.7	48.5	738
REGION OF THE AMERICAS								
2000	Antigua and Barbuda	National	100.0	91.7	100.0	91.7	55.2	1,183
2004	Antigua and Barbuda	National	100.0	91.8	100.0	91.8	48.9	1,089
2000	Argentina	Capital Federal	95.7	88.5	100.0	84.7	49.8	1,686
2003	Argentina	Capital Federal	91.7	82.0	100.0	75.2	48.0	1,295
2007	Argentina	Capital Federal	92.0	75.4	100.0	69.3	48.7	1,115
2000	Bahamas	National	92.0	75.2	100.0	69.2	53.6	1,174
2004	Bahamas	National	96.0	86.9	100.0	83.5	55.9	1,018
1999	Barbados	National	94.7	96.2	100.0	91.1	57.4	1,317
2002	Barbados	National	100.0	85.7	98.7	84.6	51.0	1,097
2007	Barbados	National	95.5	83.2	100.0	79.4	52.6	1,066
2002	Belize	National	94.1	91.9	100.0	86.5	51.6	1,016
2008	Belize	National	94.4	99.4	100.0	93.9	51.6	1,138
2000	Bolivia	La Paz	95.0	83.7	100.0	79.6	51.0	3,443
2003	Bolivia	La Paz	95.0	92.0	100.0	87.3	47.5	3,246
2002	Brazil	Curitiba	92.0	82.9	100.0	76.2	59.6	1,161
2006	Brazil	Curitiba	90.0	92.1	100.0	82.9	52.4	2,433
2000	Chile	Santiago	98.0	86.0	100.0	84.3	50.6	2,412
2003	Chile	Santiago	100.0	88.9	100.0	88.9	50.1	1,539
2008	Chile	Santiago	92.0	86.9	100.0	79.9	48.4	3,590
2001	Colombia	Bogota	100.0	92.1	100.0	92.1	49.4	2,016
2007	Colombia	Bogota	93.8	89.5	100.0	83.9	53.3	1,235
1999	Costa Rica	National	100.0	90.4	100.0	90.4	51.9	3,839
2002	Costa Rica	National	92.0	80.8	100.0	74.3	49.1	2,698
2008	Costa Rica	National	100.0	82.7	97.0	80.6	49.2	2,920

* Response rates:
Sch RR–the school response rate is calculated as the number of participating schools divided by the number of selected schools
Stu RR–the student response rate is calculated as the number of participating students divided by the number of students enrolled in a class
Class RR–the class response rate is calculated as the number of participating classes divided by the number of classes selected
Overall RR–the overall response rate is calculated as the school rate times the class rate times the student rate
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 2. (Continued)

Survey Year	WHO Region and Site	National or Subnational Sample	Response Rates*				Percent of Girls in Survey Population	No. aged 13–15 years
			Sch RR	Stu RR	Class RR	Overall RR		
2000	Cuba	Havana	100.0	91.0	100.0	91.0	47.0	1,376
2004	Cuba	Havana	100.0	91.5	100.0	91.5	50.2	1,663
2000	Dominica	National	100.0	86.6	100.0	86.6	51.5	1,004
2004	Dominica	National	100.0	89.2	100.0	89.2	54.7	990
2001	Ecuador	Quito	100.0	96.3	100.0	96.3	46.1	1,672
2007	Ecuador	Quito	92.0	95.1	100.0	87.5	52.8	1,389
2000	Grenada	National	92.5	79.2	100.0	73.3	53.4	1,807
2004	Grenada	National	100.0	89.3	100.0	89.3	55.7	1,216
2002	Guatemala	Guatemala City	84.0	80.1	100.0	67.3	60.0	1,482
2008	Guatemala	Guatemala City	96.2	85.0	100.0	81.8	54.1	2,716
2000	Guyana	National	86.0	72.1	100.0	62.0	59.8	603
2004	Guyana	National	100.0	78.6	100.0	78.6	51.3	890
2000	Haiti	Port-au-Prince	80.0	97.5	100.0	78.0	55.5	1,039
2005	Haiti	Port-au-Prince	68.0	62.2	100.0	42.3	51.7	767
2000	Jamaica	National	100.0	86.5	100.0	86.5	51.2	1,256
2006	Jamaica	National	100.0	79.9	100.0	79.9	50.5	1,398
2003	Mexico	Mexico City	100.0	92.7	100.0	92.7	49.7	1,334
2006	Mexico	Mexico City	96.0	84.5	100.0	81.2	50.2	1,412
2002	Panama	National	98.0	89.1	100.0	87.3	50.4	1,296
2008	Panama	National	96.0	83.9	100.0	80.5	51.5	2,716
2003	Paraguay	Asuncion	96.0	90.6	100.0	87.0	49.1	1,300
2008	Paraguay	Asuncion	96.0	88.4	100.0	84.9	50.0	3,275
2000	Peru	Lima	98.0	92.0	100.0	90.1	53.0	1,217
2003	Peru	Lima	96.0	90.3	100.0	86.7	49.8	1,045
2007	Peru	Lima	100.0	87.8	100.0	87.8	49.9	1,019
2000	Saint Lucia	National	100.0	86.2	100.0	86.2	58.9	1,068
2007	Saint Lucia	National	100.0	77.7	100.0	77.7	55.6	836
2000	Saint Vincent and The Grenadines	National	100.0	78.4	100.0	78.4	52.5	1,180
2007	Saint Vincent and The Grenadines	National	100.0	84.5	100.0	84.5	52.8	872
2000	Suriname	National	100.0	84.5	100.0	84.5	54.8	797
2004	Suriname	National	100.0	94.4	100.0	94.4	52.1	1,020
2000	Trinidad and Tobago	National	88.3	85.2	100.0	75.3	50.3	1,900
2007	Trinidad and Tobago	National	100.0	74.5	100.0	74.5	50.6	1,802
2000	U.S. Virgin Islands††	National	89.6	86.5	100.0	77.5	48.2	1,188
2004	U.S. Virgin Islands††	National	94.3	89.8	100.0	84.6	52.0	1,433
2000	United States of America	National	90.0	93.4	100.0	84.1	49.7	16,416

* Response rates:
Sch RR–the school response rate is calculated as the number of participating schools divided by the number of selected schools
Stu RR–the student response rate is calculated as the number of participating students divided by the number of students enrolled in a class
Class RR–the class response rate is calculated as the number of participating classes divided by the number of classes selected
Overall RR–the overall response rate is calculated as the school rate times the class rate times the student rate
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 2. (Continued)

Survey Year	WHO Region and Site	National or Subnational Sample	Response Rates*				Percent of Girls in Survey Population	No. aged 13–15 years
			Sch RR	Stu RR	Class RR	Overall RR		
2004	United States of America	National	93.0	88.0	100.0	81.8	50.4	13,515
2000	Uruguay	National	97.4	85.9	100.0	83.6	51.3	3,413
2007	Uruguay	National	100.0	74.9	100.0	74.9	55.7	2,483
1999	Venezuela	Lara State	85.7	93.3	100.0	80.0	52.8	978
2003	Venezuela	Lara State	100.0	79.3	100.0	79.3	51.8	1,307
SOUTH-EAST ASIA REGION								
2004	Bangladesh	Dhaka	100.0	55.4	100.0	55.4	28.2	1,189
2007	Bangladesh	Dhaka	100.0	89.2	100.0	89.2	51.5	1,272
2004	Bhutan	National	83.3	93.3	100.0	77.7	48.7	870
2006	Bhutan	National	93.3	95.5	100.0	89.1	50.2	1,130
2004	India	National	98.6	85.2	100.0	84.0	38.1	57,251
2006	India	National	99.4	82.3	100.0	81.8	42.1	11,789
2000	Indonesia	Jakarta	100.0	91.6	100.0	91.6	50.2	1,490
2004	Indonesia	Jakarta	96.0	93.3	100.0	89.6	48.9	1,492
2004	Maldives	National	97.2	84.2	100.0	81.9	45.9	1,689
2007	Maldives	National	97.5	87.8	100.0	85.6	52.7	1,870
1999	Sri Lanka	National	85.7	89.0	100.0	76.3	49.3	2,500
2003	Sri Lanka	National	100.0	79.1	100.0	79.1	48.4	1,667
2007	Sri Lanka	National	100.0	85.0	100.0	85.0	50.0	1,428
WESTERN PACIFIC REGION								
1999	China	Tianjin	100.0	96.8	100.0	96.8	52.8	2,437
2005	China	Tianjin	97.3	97.9	100.0	95.2	52.2	1,759
1999	Fiji	National	86.3	88.1	100.0	76.0	51.0	1,331
2005	Fiji	National	100.0	80.9	100.0	80.9	50.6	1,687
2003	Lao PDR	Vientiane Province	100.0	94.9	100.0	94.9	41.9	1,591
2007	Lao PDR	Vientiane Province	100.0	99.0	100.0	99.0	43.3	1,325
2001	Macau¶¶	Macau	90.5	94.6	100.0	85.6	46.5	1,108
2005	Macau¶¶	Macau	94.0	93.8	100.0	88.2	46.0	1,366
2003	Mongolia	National	100.0	92.6	100.0	92.6	54.9	3,521
2007	Mongolia	National	100.0	97.9	53.0	51.9	51.0	1,499
2000	Northern Mariana Islands**	National	68.8	80.3	100.0	55.2	50.1	1,308
2004	Northern Mariana Islands**	National	88.4	79.3	100.0	70.1	48.4	2,061
2000	Palau	National	100.0	90.6	100.0	90.6	50.9	822
2005	Palau	National	100.0	94.9	100.0	94.9	52.4	928
2000	Philippines	National	90.0	88.7	100.0	79.8	56.9	5,582
2004	Philippines	National	99.3	85.4	100.0	84.8	56.9	4,033
2007	Philippines	National	97.3	83.1	100.0	80.9	53.2	3,278
2003	Viet Nam	Hanoi	100.0	96.4	100.0	96.4	56.4	1,151
2007	Viet Nam	Hanoi	100.0	95.6	100.0	95.6	49.3	1,643

* Response rates:
Sch RR–the school response rate is calculated as the number of participating schools divided by the number of selected schools
Stu RR–the student response rate is calculated as the number of participating students divided by the number of students enrolled in a class
Class RR–the class response rate is calculated as the number of participating classes divided by the number of classes selected
Overall RR–the overall response rate is calculated as the school rate times the class rate times the student rate
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

surveys, 145 were national and 64 were subnational. A total of 530,849 students completed interviews for an average of 2,540 per survey. The number of students who participated in each survey varied due to the level of regional stratification included in each sample design. The school response rate ranged from 68.0% (Haiti 2005) to 100.0% in 105 of the surveys. Only 18 of the 209 surveys had school response rates less than 90%. The classroom response rate was 100% in all but 4 surveys (Barbados 2002, Costa Rica 2008, Mongolia 2007, and Swaziland 2005). The student response rate ranged from 55.4% (Bangladesh 2004) to over 80% in 190 surveys. The overall response rate was greater than 60% in 205 surveys. The exceptions with overall response rates less than 60% were Bangladesh 2004, Haiti 2005, Mongolia 2007, and Northern Mariana Islands 2000.

Results

African Region (AFR)

Current cigarette smoking

For the 20 AFR sites, current cigarette smoking decreased significantly for boys and girls in Burkina Faso, Mali, Niger, and Zimbabwe (1999 to 2003); for boys in Senegal; and for girls in Togo (Table 3). Current cigarette smoking increased significantly for boys and girls in Botswana and for girls in Mauritania. There were no significant changes over time in the rate of cigarette smoking in the remaining 12 sites.

Gender differences changed over time in 5 AFR sites. In Kenya, South Africa and Zimbabwe no gender differences were reported at Time 1, but at Time 2, boys were higher than girls. However, in Lesotho and Mauritania, the gender difference at Time 1 changed to no gender difference at Time 2; in the case of Zimbabwe the gender difference at Time 2 changed to no gender difference at Time 3 (Table 3). Gender differences in current cigarette smoking did not change over time in the remaining 15 sites: boys had significantly higher smoking than girls in 10 sites; and there were no gender differences in 5 sites.

Current other tobacco use

For the 19 AFR sites with data, use of other tobacco products decreased significantly for boys and girls in Malawi and for girls in Ghana (Table 3). Other tobacco use increased significantly for boys and girls in Botswana; for boys in Lesotho, Mozambique, and Senegal; and for girls in Swaziland. There were no significant changes over time in the rate of other tobacco use in the remaining 12 sites.

Changes in gender differences were reported in 6 AFR sites. In Nigeria, Senegal, Seychelles, and South Africa, boys had higher rates than girls at Time 1 but no gender difference at Time 2; and Burkina Faso and Togo had no gender difference at Time 1, but boys were higher than girls at Time 2 (Table 3). Gender differences in other tobacco use did not change over time in the remaining 13 sites with no gender differences in 11sites and boys had significantly higher use than girls in Mali and Swaziland.

Eastern Mediterranean Region (EMR)

Current cigarette smoking

Among the 17 EMR sites, current cigarette smoking decreased significantly for boys and girls in Jordan (2003 to 2007); for boys in Oman; and for girls in Somalia (Table 3). Smoking increased significantly for boys and girls in Syria; and girls in the West Bank and UAE. There were no significant changes in the rate of cigarette smoking over time in the remaining 11 sites.

Gender differences changed over time in 3 EMR sites. In Morocco and Yemen, gender differences at Time 1 were not reported at Time 2; and Egypt had no gender difference at Time 1, but changed to boys higher than girls at Time 2 (Table 3). Gender differences in current cigarette smoking did not change over time in the remaining 14 sites where boys had significantly higher smoking than girls in 13 sites; and there was no gender difference in Somalia.

Current other tobacco use

For the 17 EMR sites, use of other tobacco products decreased significantly for boys and girls in Egypt and Oman; and for boys in Sudan (Table 3). Other tobacco use increased significantly for boys and girls in Iran, Jordan (1999 to 2007), Tunisia, and United Arab Emirates; increased significantly for boys in Syria; and increased significantly for girls in the Gaza Strip. There were no changes over time in the rate of other tobacco use in the remaining eight sites.

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. Prevalence of tobacco use, by sex, World Health Organization (WHO) region, WHO member states, geographic region, territory, special administrative region, or commonwealth–Global Youth Tobacco Survey 1999–2008

		Current cigarette smoking				*Current use of other tobacco products*			
WHO Region and Site	*Year and Year Difference P-Value*	*Total % (95% CI)**	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*	*Total % (95% CI)*	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*
AFRICAN REGION									
Botswana	2001	8.3 (6.4–10.8)	10.4 (7.7–13.9)	6.5 (4.5–9.2)	0.0271	9.5 (7.5–11.9)	10.0 (7.2–13.8)	9.1 (6.8–12.0)	0.6143
Botswana	2008	14.3 (11.2–18.1)	18.1 (13.4–23.9)	10.9 (7.8–15.0)	0.0157	15.2 (12.8–18.0)	16.3 (13.0–20.2)	14.3 (11.7–17.5)	0.3129
Botswana	P-Value	0.0037	0.0114	0.0386		0.0010	0.0107	0.0073	
Burkina Faso (Ouagadougou)	2001	19.0 (14.7–24.1)	26.1 (19.5–33.9)	11.6 (8.5–15.6)	0.0000	6.9 (4.6–10.4)	7.9 (4.9–12.4)	6.3 (4.0–9.6)	0.2539
Burkina Faso (Ouagadougou)	2006	8.4 (6.3–11.1)	14.1 (10.4–18.7)	2.4 (1.3–4.3)	0.0000	7.2 (5.6–9.1)	9.3 (7.1–12.1)	4.8 (3.0–7.5)	0.0080
Burkina Faso (Ouagadougou)	P-Value	0.0002	0.0044	0.0000		0.8824	0.5042	0.3712	
Ghana	2000	4.2 (2.7–6.4)	5.0 (3.0–8.2)	3.1 (1.7–5.7)	0.1933	14.6 (11.3–18.5)	13.6 (9.9–18.3)	15.5 (11.4–20.6)	0.4064
Ghana	2006	2.7 (1.9–4.0)	2.8 (1.7–4.7)	2.3 (1.4–3.5)	0.4787	10.4 (7.8–13.7)	10.1 (7.3–13.8)	10.1 (7.6–13.2)	0.8952
Ghana	P-Value	0.1632	0.1251	0.4303		0.0705	0.1826	0.0421	
Kenya	2001	6.6 (4.5–9.5)	8.7 (5.9–12.6)	4.7 (2.4–8.7)	0.0547	8.9 (6.7–11.9)	9.0 (6.2–12.7)	8.9 (6.3–12.3)	0.9473
Kenya	2007	8.2 (6.1–11.1)	11.2 (8.9–14.0)	5.2 (3.5–7.6)	0.0000	10.1 (6.5–15.4)	8.2 (6.1–11.0)	11.4 (5.2–23.2)	0.5206
Kenya	P-Value	0.3424	0.2147	0.7742		0.6435	0.6969	0.5686	
Lesotho	2002	9.2 (6.6–12.6)	16.6 (12.4–21.9)	4.8 (3.4–6.9)	0.0000	14.8 (12.7–17.2)	12.3 (9.5–15.7)	14.8 (12.6–17.3)	0.1581
Lesotho	2008	10.1 (6.9–14.4)	11.8 (7.0–19.3)	7.5 (4.9–11.2)	0.0978	19.5 (16.1–23.4)	20.4 (15.2–26.9)	17.9 (14.6–21.8)	0.3810

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Lesotho	P-Value	0.6997	0.2041	0.1314		0.028	0.0137	0.1406	
Malawi (Blantyre)	2000	2.4 (0.8–6.6)	4.2 (1.1–14.1)	0.9 (0.5–1.7)	0.2238	14.7 (11.9–18.1)	14.4 (10.1–20.0)	15.2 (11.4–19.9)	0.7995
Malawi (Blantyre)	2005	1.7 (0.8–3.8)	2.0 (1.0–4.1)	1.4 (0.3–5.7)	0.5245	8.6 (6.7–10.9)	8.8 (6.3–12.1)	8.3 (5.7–12.0)	0.8057
Malawi (Blantyre)	P-Value	0.6087	0.3995	0.5688		0.0014	0.0436	0.0090	
Mali (Bamako)	2001	23.5 (19.6–27.9)	41.8 (34.0–50.0)	4.6 (2.7–7.7)	0.0000	9.1 (7.0–11.7)	13.1 (9.5–17.9)	4.8 (2.6–8.4)	0.0066
Mali (Bamako)	2008	9.5 (6.8–13.1)	18.6 (12.1–27.5)	1.7 (0.8–3.9)	0.0004	9.1 (6.1–13.2)	14.8 (9.5–22.2)	4.2 (2.6–6.8)	0.0018
Mali (Bamako)	P-Value	0.0000	0.0001	0.0332		1	0.6441	0.7231	
Mauritania	2001	14.8 (12.2–17.8)	19.8 (16.2–24.1)	8.8 (6.6–11.6)	0.0000	14.9 (13.0–17.0)	15.8 (13.3–18.8)	13.4 (10.5–16.9)	0.2725
Mauritania	2006	19.5 (16.3–23.2)	20.3 (17.5–23.4)	18.3 (13.4–24.5)	0.4886	18.0 (14.4–22.2)	18.4 (14.3–23.4)	17.3 (12.1–24.1)	0.7449
Mauritania	P-Value	0.0369	0.8540	0.0021		0.1629	0.3225	0.2449	
Mauritius	2003	13.2 (8.6–19.8)	19.8 (14.4–26.5)	7.7 (4.3–13.4)	0.0019	NA	NA	NA	NA
Mauritius	2008	13.7 (9.3–19.8)	20.3 (13.9–28.6)	7.7 (4.1–14.0)	0.0078	NA	NA	NA	NA
Mauritius	P-Value	0.8838	0.9154	1		NA	NA	NA	
Mozambique (Maputo City)	2002	3.1 (2.0–4.7)	5.0 (2.9–8.5)	1.4 (0.6–3.3)	0.0208	5.8 (4.7–7.2)	5.4 (3.9–7.5)	6.0 (4.5–8.0)	0.6363
Mozambique (Maputo City)	2007	2.7 (1.6–4.7)	4.5 (2.6–7.9)	1.2 (0.4–3.5)	0.0128	8.2 (6.2–10.7)	9.6 (6.8–13.3)	6.8 (4.3–10.4)	0.1961
Mozambique (Maputo City)	P-Value	0.739	0.7941	0.8372		0.056	0.0221	0.6507	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking: Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Current use of other tobacco products: Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Niger	2001	14.8 (11.5–18.8)	22.4 (17.8–27.8)	6.1 (3.0–11.8)	0.0001	7.6 (5.3–10.9)	6.7 (4.5–9.8)	7.5 (4.6–12.0)	0.5928
Niger	2006	6.3 (4.2–9.2)	11.7 (7.6–17.4)	1.1 (0.3–3.9)	0.0005	6.6 (4.4–9.6)	6.1 (3.8–9.5)	7.0 (4.6–10.5)	0.4041
Niger	P-Value	0.0002	0.0025	0.0238		0.5433	0.7429	0.8172	
Nigeria (Cross River State)	2000	7.0 (4.5–10.9)	7.7 (4.4–13.3)	3.3 (1.9–5.8)	0.0383	14.0 (11.0–17.7)	18.6 (14.2–24.0)	9.4 (6.5–13.5)	0.0036
Nigeria (Cross River State)	2008	4.1 (1.4–11.1)	6.8 (2.4–17.7)	1.2 (0.2–6.4)	0.0468	23.3 (16.6–31.6)	23.9 (16.2–33.9)	17.5 (9.9–29.0)	0.3280
Nigeria (Cross River State)	P-Value	0.2294	0.7982	0.1030		0.0190	0.2555	0.0875	
Senegal	2002	13.2 (10.9–15.8)	20.2 (16.4–24.7)	4.4 (3.0–6.6)	0.0000	5.4 (4.1–7.0)	7.3 (5.5–9.8)	2.9 (2.0–4.2)	0.0002
Senegal	2007	7.5 (4.6–12.1)	12.1 (7.6–18.9)	2.7 (1.3–5.4)	0.0005	9.3 (5.5–15.3)	11.7 (8.4–15.9)	7.7 (3.0–18.0)	0.1004
Senegal	P-Value	0.0102	0.0199	0.1666		0.1104	0.0420	0.1638	
Seychelles	2002	26.8 (21.9–32.2)	29.9 (23.3–37.4)	23.9 (18.7–30.0)	0.1356	9.3 (7.0–12.3)	13.0 (9.5–17.4)	5.5 (3.5–8.4)	0.0007
Seychelles	2007	21.5 (16.7–27.2)	23.2 (17.4–30.2)	20.0 (15.0–26.2)	0.2900	10.5 (7.7–14.1)	10.6 (6.9–16.0)	9.2 (6.4–13.0)	0.5849
Seychelles	P-Value	0.1518	0.1629	0.3266		0.5724	0.4256	0.0669	
South Africa	1999	17.6 (15.2–20.3)	20.0 (16.0–24.6)	15.3 (11.5–20.1)	0.1701	11.8 (8.8–15.7)	15.7 (10.5–22.9)	9.4 (6.4–13.5)	0.0397
South Africa	2002	14.8 (12.9–17.0)	21.0 (16.7–26.1)	10.6 (8.3–13.4)	0.0005	13.1 (11.6–14.7)	14.8 (12.6–17.3)	11.9 (10.1–14.0)	0.0637
South Africa	P-Value	0.0884	0.7469	0.0621		0.5039	0.7761	0.1999	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Swaziland	2001	6.4 (5.2–7.9)	10.4 (8.2–13.2)	3.4 (2.5–4.6)	0.0000	6.8 (5.8–8.0)	8.9 (6.9–11.4)	5.2 (4.4–6.2)	0.0027
Swaziland	2005	5.6 (4.9–6.4)	8.9 (7.8–10.2)	3.2 (2.5–4.2)	0.0000	7.5 (6.5–8.7)	8.5 (7.1–10.1)	6.9 (5.8–8.2)	0.0222
Swaziland	P-Value	0.2884	0.2763	0.7800		0.3586	0.7629	0.0295	
Togo	2002	11.0 (8.2–14.5)	14.9 (11.1–19.6)	4.0 (2.6–6.3)	0.0000	8.7 (6.8–11.0)	9.5 (7.2–12.5)	7.1 (5.2–9.6)	0.1069
Togo	2007	6.2 (3.6–10.2)	9.1 (5.1–15.6)	1.7 (1.1–2.6)	0.0066	10.4 (8.5–12.8)	12.1 (9.3–15.7)	7.4 (5.2–10.4)	0.0222
Togo	P-Value	0.0304	0.0798	0.0161		0.2277	0.2022	0.8667	
Uganda (Kampala)	2002	3.2 (1.7–5.6)	3.7 (1.6–8.3)	2.6 (1.2–5.4)	0.4847	9.7 (7.8–12.1)	9.7 (8.5–11.2)	9.8 (6.8–13.9)	0.9952
Uganda (Kampala)	2007	5.2 (3.5–7.7)	5.7 (3.5–9.2)	4.2 (2.7–6.4)	0.2492	13.3 (10.4–16.9)	14.3 (10.2–19.7)	12.0 (8.8–16.1)	0.3293
Uganda (Kampala)	P-Value	0.1232	0.2888	0.1948		0.0566	0.0575	0.3418	
Zambia (Lusaka)	2002	9.2 (6.7–12.6)	9.4 (6.3–13.9)	8.7 (4.6–15.9)	0.8268	17.7 (14.0–22.0)	17.1 (12.7–22.8)	17.3 (12.2–23.9)	0.9690
Zambia (Lusaka)	2007	6.8 (4.3–10.5)	6.7 (4.0–11.1)	6.8 (4.0–11.3)	0.9884	22.8 (17.3–29.4)	22.8 (16.7–30.2)	22.8 (16.9–30.1)	0.9851
Zambia (Lusaka)	P-Value	0.2210	0.2599	0.5276		0.1369	0.1580	0.1856	
Zimbabwe (Harare)	1999	10.7 (7.4–15.2)	11.6 (8.7–15.4)	9.9 (5.6–17.1)	0.5538	9.5 (6.4–14.0)	11.0 (7.2–16.5)	8.4 (4.6–14.8)	0.3674
Zimbabwe (Harare)	2003	4.7 (3.1–7.0)	6.1 (4.0–9.4)	3.2 (1.8–5.7)	0.0150	6.6 (4.5–9.6)	8.4 (5.2–13.5)	4.8 (3.4–6.8)	0.0767
Zimbabwe (Harare)	2008	3.2 (1.7–5.7)	4.8 (2.6–9.0)	1.5 (0.5–4.6)	0.0567	9.6 (6.9–13.3)	10.9 (7.7–15.1)	7.5 (4.8–11.6)	0.1154

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking: Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Current use of other tobacco products: Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Zimbabwe (Harare)	P-Value 99, 03	0.0052	0.0103	0.0198		0.1673	0.3768	0.1504	
Zimbabwe (Harare)	P-Value 03, 08	0.2287	0.4874	0.1492		0.1167	0.3406	0.1297	
EASTERN MEDITERRANEAN REGION									
Egypt	2001	4.2 (2.8–6.1)	3.9 (2.7–5.5)	4.0 (2.1–7.6)	0.9123	15.3 (12.5–18.7)	18.3 (14.2–23.4)	12.0 (8.4–16.7)	0.0667
Egypt	2005	4.0 (2.7–5.8)	5.9 (4.4–7.9)	1.4 (0.9–2.3)	0.0000	10.1 (8.1–12.4)	12.3 (9.5–15.8)	6.7 (5.1–8.6)	0.0049
Egypt	P-Value	0.8964	0.0666	0.0545		0.0060	0.0311	0.0173	
Gaza Strip§§	2000	9.0 (6.4–12.5)	15.1 (9.8–22.4)	3.4 (2.2–5.3)	0.0018	7.8 (5.5–10.9)	12.4 (8.4–17.8)	3.6 (2.1–6.0)	0.0020
Gaza Strip§§	2005	6.6 (3.9–10.9)	9.7 (6.3–14.6)	3.0 (1.6–5.4)	0.0059	11.7 (9.1–14.9)	12.8 (9.3–17.2)	10.0 (6.9–14.3)	0.3131
Gaza Strip§§	P-Value	0.2703	0.1287	0.7107		0.0408	0.8836	0.0024	
Iran	2003	2.0 (1.4–2.8)	3.2 (2.3–4.5)	1.0 (0.6–1.6)	0.0003	12.1 (10.2–14.4)	16.0 (13.0–19.5)	8.7 (6.9–11.0)	0.0006
Iran	2007	3.0 (1.7–5.5)	5.1 (2.8–9.1)	0.9 (0.4–1.9)	0.0122	26.1 (20.6–32.4)	31.9 (24.4–40.4)	19.5 (15.6–24.2)	0.0061
Iran	P-Value	0.1855	0.1120	0.3885		0.0001	0.0001	0.0182	
Jordan	1999	16.6 (13.8–19.8)	22.0 (18.2–26.4)	9.9 (7.8–12.6)	0.0000	11.2 (9.1–13.6)	14.5 (11.4–18.3)	7.1 (5.7–8.8)	0.0002
Jordan	2003	17.7 (14.7–21.3)	21.4 (17.9–25.4)	12.6 (10.1–15.7)	0.0001	20.0 (18.2–21.8)	21.4 (19.0–24.0)	18.2 (16.3–20.3)	0.0349

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Jordan	2007	10.3 (7.9–13.3)	13.2 (9.9–17.5)	7.1 (4.9–10.3)	0.0062	26.4 (23.8–29.1)	28.2 (24.4–32.3)	23.5 (20.8–26.5)	0.0372
Jordan	P-Value 99, 03	0.5921	0.8285	0.1455		0.0000	0.0016	0.0000	
Jordan	P-Value 03, 07	0.0006	0.0026	0.0049		0.0001	0.0042	0.0027	
Kuwait	2001	10.0 (8.5–11.7)	14.8 (12.8–17.0)	4.9 (3.9–6.2)	0.0000	16.2 (14.7–17.9)	19.1 (17.1–21.4)	12.9 (11.3–14.7)	0.0000
Kuwait	2005	10.8 (7.7–15.1)	17.7 (14.2–21.7)	4.5 (3.0–6.9)	0.0000	14.5 (12.3–16.9)	17.4 (15.0–20.1)	11.7 (9.9–13.9)	0.0002
Kuwait	P-Value	0.6603	0.1754	0.7036		0.2077	0.2981	0.3587	
Lebanon	2001	7.5 (6.1–9.2)	10.4 (7.6–14.1)	5.3 (3.9–7.2)	0.0130	38.6 (35.3–42.0)	45.0 (39.9–50.2)	33.9 (30.2–37.8)	0.0006
Lebanon	2005	8.6 (6.8–10.8)	11.8 (8.5–16.3)	5.6 (4.2–7.5)	0.0061	40.0 (37.8–42.3)	44.7 (40.8–48.6)	35.7 (32.5–39.1)	0.0031
Lebanon	P-Value	0.3686	0.5647	0.7573		0.4714	0.9163	0.4525	
Libya	2003	4.1 (2.6–6.5)	7.3 (4.5–11.6)	0.8 (0.3–2.4)	0.0005	9.8 (7.4–12.8)	11.7 (8.1–16.4)	7.5 (5.1–10.8)	0.0586
Libya	2007	4.6 (2.9–7.2)	7.7 (4.9–11.9)	0.9 (0.3–2.5)	0.0001	7.2 (5.4–9.5)	8.6 (5.2–14.0)	5.6 (4.1–7.7)	0.2326
Libya	P-Value	0.7285	0.8540	0.9515		0.1140	0.3006	0.2539	
Morocco	2001	2.6 (1.9–3.4)	3.9 (2.9–5.3)	1.0 (0.4–2.2)	0.0006	9.2 (8.1–10.5)	10.4 (9.0–11.9)	7.6 (6.0–9.7)	0.0190

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

		Current cigarette smoking				*Current use of other tobacco products*			
WHO Region and Site	*Year and Year Difference P-Value*	*Total % (95% CI)**	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*	*Total % (95% CI)*	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*
Morocco	2006	3.5 (2.7–4.6)	4.3 (2.9–6.4)	2.1 (1.1–3.9)	0.0516	9.0 (7.5–10.8)	10.3 (7.8–13.5)	6.9 (5.5–8.7)	0.0304
Morocco	P-Value	0.1062	0.6656	0.1533		0.8541	0.9574	0.5346	
Oman	2002	6.8 (3.3–13.5)	14.2 (6.5–28.1)	1.8 (1.0–3.4)	0.0245	9.4 (7.3–12.1)	14.0 (10.7–18.2)	6.4 (4.6–8.7)	0.0010
Oman	2007	2.3 (1.1–4.8)	3.5 (1.8–6.6)	1.2 (0.3–4.1)	0.0450	5.7 (3.5–9.1)	7.1 (4.2–11.9)	3.3 (1.7–6.5)	0.0428
Oman	P-Value	0.0740	0.0461	0.4927		0.0370	0.0096	0.0478	
Qatar	2004	6.4 (5.3–7.8)	10.7 (8.8–13.0)	2.8 (1.7–4.8)	0.0000	13.7 (12.0–15.7)	15.9 (13.1–19.1)	12.0 (9.9–14.5)	0.0353
Qatar	2007	6.5 (4.7–8.9)	13.4 (9.5–18.7)	2.3 (1.0–5.1)	0.0001	15.6 (13.1–18.6)	19.4 (15.7–23.8)	12.6 (9.3–16.8)	0.0167
Qatar	P-Value	0.9357	0.2664	0.6447		0.2483	0.1504	0.7744	
Somalia (Somaliland)	2004	18.6 (10.9–29.8)	8.6 (3.6–19.3)	14.8 (7.4–27.5)	0.0809	18.7 (11.1–29.6)	15.0 (7.4–28.1)	15.0 (7.3–28.4)	1.0000
Somalia (Somaliland)	2007	5.8 (4.0–8.4)	4.9 (3.2–7.4)	4.5 (1.6–11.8)	0.8736	12.5 (10.1–15.4)	12.7 (10.2–15.8)	9.8 (5.2–17.6)	0.3502
Somalia (Somaliland)	P-Value	0.0075	0.2885	0.0467		0.1707	0.6399	0.3470	
Sudan	2001	6.1 (3.8–9.8)	10.8 (6.3–17.8)	1.9 (1.1–3.5)	0.0052	13.5 (11.2–16.2)	17.2 (13.9–21.0)	10.4 (8.0–13.5)	0.0028
Sudan	2005	6.0 (3.6–10.0)	10.2 (6.6–15.5)	2.1 (1.4–3.2)	0.0016	10.2 (8.0–12.9)	11.0 (7.8–15.4)	9.3 (7.0–12.2)	0.4064
Sudan	P-Value	0.9576	0.8732	0.7863		0.0515	0.0155	0.5213	
Syria	2002	6.3 (4.8–8.1)	8.1 (5.7–11.4)	3.1 (1.8–5.4)	0.0094	17.6 (15.0–20.6)	19.2 (16.0–22.9)	14.5 (11.0–18.9)	0.0579

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Syria	2007	12.3 (9.3–16.1)	19.1 (14.6–24.7)	5.9 (4.3–8.2)	0.0000	22.6 (19.1–26.4)	29.7 (25.1–34.7)	15.3 (12.4–18.9)	0.0000
Syria	P-Value	0.0016	0.0003	0.0305		0.0323	0.0007	0.7455	
Tunisia	2001	10.1 (8.5–11.8)	17.6 (14.7–21.0)	3.0 (2.1–4.4)	0.0000	7.2 (5.9–8.7)	11.3 (9.2–13.9)	3.1 (2.4–4.1)	0.0000
Tunisia	2007	8.3 (6.6–10.4)	15.1 (12.3–18.4)	1.6 (0.8–3.1)	0.0000	13.9 (11.6–16.5)	19.9 (16.1–24.3)	7.8 (5.8–10.4)	0.0000
Tunisia	P-Value	0.1474	0.2388	0.0591		0.0000	0.0005	0.0003	
United Arab Emirates	2002	6.8 (5.2–8.9)	11.7 (8.9–15.4)	2.2 (1.5–3.4)	0.0000	15.0 (13.0–17.3)	19.0 (16.1–22.3)	10.2 (8.1–12.9)	0.0001
United Arab Emirates	2005	8.0 (6.6–9.7)	12.1 (10.3–14.1)	3.6 (2.9–4.4)	0.0000	28.8 (26.7–30.9)	32.7 (30.4–35.1)	24.7 (21.9–27.7)	0.0000
United Arab Emirates	P-Value	0.3158	0.8470	0.0271		0.0000	0.0000	0.0000	
West Bank§§	2000	14.2 (11.2–17.7)	24.7 (21.5–28.3)	4.7 (3.8–6.0)	0.0000	16.6 (14.7–18.7)	23.7 (20.8–26.8)	10.0 (8.7–11.6)	0.0000
West Bank§§	2005	18.0 (12.5–25.3)	27.6 (21.3–35.1)	8.7 (5.8–12.8)	0.0005	16.7 (14.2–19.5)	20.8 (17.2–24.8)	12.7 (9.8–16.3)	0.0161
West Bank§§	P-Value	0.3305	0.5523	0.0185		0.2483	0.7072	0.1016	
Yemen	2003	5.3 (4.4–6.3)	6.5 (5.5–7.6)	3.0 (1.9–4.5)	0.0000	14.6 (13.5–15.8)	15.7 (14.4–17.2)	12.1 (10.5–13.9)	0.0007
Yemen	2008	3.9 (2.5–6.2)	4.2 (2.3–7.5)	1.6 (0.8–3.1)	0.0591	12.4 (8.7–17.4)	12.1 (7.2–19.6)	10.1 (5.5–18.0)	0.5976
Yemen	P-Value	0.1622	0.0675	0.0939		0.3042	0.2184	0.5048	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

		Current cigarette smoking				*Current use of other tobacco products*			
WHO Region and Site)	*Year and Year Difference P-Value*	*Total % (95% CI)**	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*	*Total % (95% CI)*	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*
EUROPEAN REGION									
Bosnia and Herzegovina	2003	11.5 (9.6–13.6)	13.6 (11.0–16.8)	8.9 (7.0–11.2)	0.0004	6.4 (5.5–7.4)	7.6 (6.1–9.5)	4.3 (3.3–5.4)	0.0021
Bosnia and Herzegovina	2008	11.7 (9.9–13.9)	14.3 (12.3–16.6)	9.4 (7.3–12.0)	0.0000	7.5 (6.4–8.8)	9.3 (7.9–10.8)	5.8 (4.4–7.6)	0.0005
Bosnia and Herzegovina	P-Value	0.8541	0.7122	0.7568		0.1333	0.1452	0.1068	
Bulgaria	2002	33.1 (30.1–36.3)	26.0 (21.6–30.8)	39.4 (34.2–44.9)	0.0013	4.3 (3.1–5.9)	5.3 (3.6–7.7)	3.3 (2.2–5.1)	0.0571
Bulgaria	2008	28.2 (24.1–32.7)	24.4 (20.2–29.2)	31.6 (25.9–37.9)	0.0207	8.8 (7.0–10.9)	10.5 (7.8–13.9)	6.8 (5.6–8.3)	0.0179
Bulgaria	P-Value	0.0629	0.6340	0.0529		0.0003	0.0050	0.0007	
Croatia	2003	18.3 (15.4–21.6)	19.9 (16.1–24.3)	16.4 (13.4–19.9)	0.0730	10.2 (8.4–12.4)	11.1 (9.0–13.6)	9.2 (6.7–12.4)	0.2188
Croatia	2007	24.1 (19.9–28.7)	21.7 (17.9–26.0)	25.6 (20.6–31.2)	0.0901	13.9 (11.9–16.2)	14.6 (13.1–16.2)	13.0 (10.0–16.9)	0.3065
Croatia	P-Value	0.0318	0.5302	0.0037		0.0124	0.0112	0.0808	
Czech Republic	2002	34.6 (31.2–38.1)	34.0 (29.7–38.5)	35.1 (30.8–39.6)	0.6700	8.5 (7.1–10.1)	11.6 (9.6–14.1)	5.3 (3.9–7.1)	0.0000
Czech Republic	2007	31.1 (27.2–35.3)	29.8 (25.1–35.0)	32.7 (27.6–38.1)	0.3489	14.5 (12.0–17.3)	17.2 (14.3–20.7)	11.2 (8.4–15.0)	0.0048
Czech Republic	P-Value	0.1938	0.2045	0.4692		0.0001	0.0051	0.0015	
Estonia	2003	28.9 (25.8–32.3)	29.8 (26.6–33.2)	27.4 (24.1–30.9)	0.1153	14.1 (12.1–16.3)	15.5 (13.4–18.0)	12.2 (10.3–14.5)	0.0065
Estonia	2007	27.2 (23.5–31.2)	28.2 (23.5–33.3)	26.2 (21.6–31.4)	0.5119	21.1 (18.5–23.9)	25.4 (21.6–29.7)	16.7 (13.6–20.4)	0.0021

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Estonia	P-Value	0.4796	0.5691	0.687		0.0001	0.0001	0.0244	
Georgia	2003	23.7 (21.0–26.7)	35.5 (30.9–40.3)	12.9 (10.2–16.2)	0.0000	6.5 (5.5–7.8)	9.6 (7.6–12.0)	3.7 (2.8–4.9)	0.0000
Georgia	2008	8.6 (5.5–13.2)	15.2 (9.9–22.8)	2.8 (1.0–7.8)	0.0000	NA	NA	NA	NA
Georgia	P-Value	0.0000	0.0000	0.0000		NA	NA	NA	
Hungary	2003	27.2 (24.6–30.1)	26.7 (22.7–31.2)	26.8 (22.9–31.2)	0.9740	5.5 (4.2–7.1)	8.2 (6.0–11.0)	3.0 (2.3–3.9)	0.0000
Hungary	2008	23.2 (19.2–27.7)	21.5 (16.6–27.4)	23.6 (19.4–28.3)	0.4492	13.8 (11.4–16.5)	16.8 (13.2–21.0)	10.4 (8.0–13.3)	0.0023
Hungary	P-Value	0.1061	0.1307	0.2800		0.0000	0.0003	0.0000	
Kyrgyzstan	2004	5.5 (4.2–7.2)	7.6 (5.6–10.2)	4.2 (3.0–5.8)	0.0023	4.8 (3.4–6.9)	7.6 (4.8–11.9)	2.9 (1.7–5.0)	0.0154
Kyrgyzstan	2008	4.4 (3.3–5.7)	6.8 (5.0–9.4)	2.2 (1.4–3.6)	0.0004	5.5 (3.9–7.7)	7.3 (5.1–10.4)	3.8 (2.5–5.8)	0.0016
Kyrgyzstan	P-Value	0.2258	0.6490	0.0250		0.6125	0.8608	0.4130	
Latvia	2002	31.1 (28.0–34.3)	33.8 (30.7–37.1)	27.8 (23.5–32.5)	0.0097	10.4 (8.6–12.5)	13.7 (11.0–17.1)	7.0 (5.4–9.0)	0.0002
Latvia	2007	32.9 (27.2–39.0)	36.3 (30.9–42.1)	30.2 (24.1–37.0)	0.0012	37.5 (32.8–42.5)	42.0 (36.1–48.1)	33.6 (29.2–38.4)	0.0018
Latvia	P-Value	0.5871	0.4347	0.5393		0.0000	0.0000	0.0000	
Lithuania	2001	33.5 (29.2–38.1)	37.6 (32.4–43.1)	29.6 (24.4–35.4)	0.0158	7.0 (5.2–9.4)	9.3 (7.2–12.0)	4.9 (3.0–7.9)	0.0004
Lithuania	2005	29.6 (26.5–32.8)	33.8 (29.4–38.6)	25.9 (21.2–31.2)	0.0368	9.1 (7.1–11.6)	13.2 (9.9–17.2)	5.7 (3.7–8.7)	0.0010
Lithuania	P-Value	0.1479	0.2808	0.3096		0.1597	0.0755	0.6229	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

		Current cigarette smoking				*Current use of other tobacco products*			
WHO Region and Site	*Year and Year Difference P-Value*	*Total % (95% CI)**	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*	*Total % (95% CI)*	*Boy % (95% CI)*	*Girl % (95% CI)*	*Boy vs Girl P-Value*
Montenegro	2004	5.6 (4.1–7.7)	6.0 (4.2–8.6)	5.0 (3.2–7.6)	0.4280	4.1 (3.2–5.3)	4.1 (2.9–5.6)	4.1 (2.8–5.8)	0.0073
Montenegro	2008	5.1 (4.0–6.4)	5.7 (4.3–7.6)	4.4 (3.1–6.1)	0.5988	3.6 (2.9–4.5)	3.7 (2.6–5.2)	3.5 (2.6–4.8)	0.0763
Montenegro	P-Value	0.3496	0.2862	0.6455		0.6790	0.7944	0.6457	
Poland (Urban)	1999	25.0 (21.8–28.5)	26.3 (21.5–31.7)	23.3 (18.9–28.4)	0.0000	11.4 (8.8–14.7)	15.4 (11.1–21.0)	8.0 (5.8–11.0)	0.0000
Poland (Urban)	2003	21.2 (16.5–26.7)	22.1 (14.9–31.6)	20.1 (16.0–24.9)	0.0000	8.7 (6.5–11.7)	11.2 (7.2–16.9)	6.4 (4.4–9.2)	0.0001
Poland (Urban)	P-Value	0.1954	0.3767	0.2992		0.1682	0.2088	0.3448	
Republic of Moldova	2004	13.7 (11.0–16.9)	23.0 (18.5–28.2)	6.0 (4.4–8.2)	0.3664	8.3 (6.4–10.5)	12.8 (9.5–17.0)	4.2 (3.2–5.6)	1.0000
Republic of Moldova	2008	11.3 (9.3–13.7)	18.5 (15.0–22.6)	5.6 (4.3–7.2)	0.1772	8.2 (6.6–10.1)	11.6 (9.0–14.8)	5.1 (3.7–7.0)	0.8442
Republic of Moldova	P-Value	0.1952	0.1374	0.7040		0.9463	0.6068	0.3705	
Russian Federation (Moscow)	1999	33.4 (30.5–36.3)	38.5 (35.0–42.1)	28.4 (25.1–32.1)	0.0000	10.7 (9.4–12.0)	15.4 (13.2–17.9)	6.0 (5.1–7.0)	0.0000
Russian Federation (Moscow)	2004	25.2 (20.8–30.1)	23.3 (17.2–30.9)	27.2 (20.5–35.2)	0.4241	15.6 (12.9–18.8)	18.4 (14.5–23.1)	12.5 (8.4–18.2)	0.0945
Russian Federation (Moscow)	P-Value	0.0016	0.0001	0.7392		0.0009	0.1871	0.0036	
Serbia	2003	12.8 (10.8–15.2)	12.2 (10.1–14.6)	13.1 (10.5–16.2)	0.5105	6.0 (5.0–7.3)	5.3 (4.1–6.7)	6.7 (5.0–8.9)	0.2501
Serbia	2008	9.3 (6.9–12.5)	9.3 (6.3–13.4)	8.9 (6.6–11.9)	0.7542	5.8 (4.7–7.2)	5.5 (3.8–8.0)	5.8 (4.4–7.6)	0.8440
Serbia	P-Value	0.0488	0.1620	0.0312		0.7900	0.8052	0.4659	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Slovakia	2003	26.4 (24.3–28.7)	28.1 (25.1–31.4)	24.3 (22.0–26.8)	0.0266	13.1 (11.6–14.7)	14.7 (12.4–17.5)	11.2 (9.6–13.0)	0.0210
Slovakia	2007	25.0 (22.6–27.6)	26.5 (23.2–29.9)	23.4 (20.7–26.4)	0.1057	12.9 (11.6–14.4)	15.1 (13.5–16.9)	10.6 (9.1–12.3)	0.0001
Slovakia	P-Value	0.3917	0.4569	0.6280		0.8649	0.7943	0.5775	
Slovenia	2003	23.2 (20.1–26.6)	21.4 (18.4–24.7)	23.9 (19.9–28.5)	0.2097	7.6 (6.3–9.2)	10.3 (8.5–12.4)	5.2 (4.0–6.9)	0.0000
Slovenia	2007	20.3 (16.3–24.9)	15.2 (10.7–21.2)	23.0 (18.7–27.9)	0.0240	8.4 (5.8–12.0)	8.3 (5.3–12.7)	7.4 (5.0–10.7)	0.5908
Slovenia	P-Value	0.2650	0.0419	0.7550		0.6368	0.3173	0.1712	
Ukraine (Kiev)	1999	33.9 (30.8–37.1)	37.8 (33.5–42.4)	29.5 (25.9–33.3)	0.0025	6.8 (5.7–8.1)	8.1 (6.6–9.9)	5.3 (4.0–7.1)	0.0102
Ukraine (Kiev)	2005	29.0 (23.1–35.6)	29.9 (22.5–38.4)	28.0 (20.3–37.2)	0.7050	15.7 (8.4–27.4)	17.8 (8.8–32.9)	13.2 (7.5–22.2)	0.1360
Ukraine (Kiev)	P-Value	0.135	0.0635	0.7277		0.0411	0.0772	0.0222	
REGION OF THE AMERICAS									
Antigua and Barbuda	2000	4.9 (3.5–6.7)	5.2 (3.3–8.0)	4.2 (2.7–6.5)	0.5114	9.6 (7.5–12.0)	10.8 (7.9–14.5)	8.4 (5.9–12.0)	0.2636
Antigua and Barbuda	2004	3.6 (2.4–5.4)	2.7 (1.7–4.3)	4.4 (2.3–8.2)	0.2864	12.4 (9.9–15.4)	13.4 (10.1–17.5)	10.9 (8.0–14.7)	0.2730
Antigua and Barbuda	P-Value	0.2283	0.0585	0.9208		0.1084	0.2910	0.2584	
Argentina (Capital Federal)	2000	25.3 (21.7–29.3)	21.9 (18.2–26.2)	28.1 (23.3–33.5)	0.0340	7.0 (6.0–8.3)	8.8 (7.2–10.9)	5.2 (3.7–7.2)	0.0136
Argentina (Capital Federal)	2003	21.9 (18.3–26.0)	17.2 (13.3–22.0)	26.8 (22.1–32.1)	0.0014	6.2 (4.4–8.8)	9.2 (6.8–12.4)	3.2 (1.6–6.1)	0.0007

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking: Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Current use of other tobacco products: Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Argentina (Capital Federal)	2007	18.5 (13.9–24.3)	15.1 (11.2–20.2)	21.4 (15.2–29.3)	0.0411	9.6 (7.2–12.6)	11.6 (8.5–15.7)	7.5 (5.3–10.5)	0.0170
Argentina (Capital Federal)	P-Value 00, 03	0.1961	0.0994	0.7010		0.5030	0.8002	0.1259	
Argentina (Capital Federal)	P-Value 03, 07	0.2646	0.4698	0.1864		0.0450	0.2587	0.0085	
Bahamas	2000	7.1 (5.3–9.3)	8.6 (6.0–12.3)	5.3 (3.6–7.8)	0.0517	11.8 (9.7–14.2)	14.7 (10.8–19.7)	9.3 (7.4–11.7)	0.0421
Bahamas	2004	5.2 (4.0–6.7)	6.2 (3.8–10.1)	3.7 (2.1–6.6)	0.2656	8.4 (6.9–10.3)	9.4 (7.4–11.8)	7.4 (5.5–9.9)	0.1477
Bahamas	P-Value	0.1227	0.2584	0.2789		0.0192	0.0338	0.2040	
Barbados	1999	10.8 (7.1–16.0)	9.2 (5.7–14.3)	11.9 (7.8–17.6)	0.1445	9.0 (6.8–11.7)	10.1 (7.1–14.1)	8.4 (5.9–11.8)	0.3601
Barbados	2002	7.0 (5.6–8.7)	7.6 (5.5–10.4)	6.4 (4.3–9.4)	0.5261	10.3 (7.8–13.4)	11.9 (8.2–16.8)	8.7 (6.8–11.0)	0.0678
Barbados	2007	11.6 (8.9–15.0)	14.3 (10.4–19.3)	9.3 (6.4–13.2)	0.0474	24.2 (21.3–27.3)	30.2 (26.1–34.6)	18.7 (15.6–22.1)	0.0000
Barbados	P-Value 99, 02	0.0875	0.5007	0.0406		0.4661	0.4907	0.8653	
Barbados	P-Value 02, 07	0.0072	0.0087	0.1691		0.0000	0.0000	0.0000	
Belize	2002	14.7 (11.6–18.5)	18.9 (15.0–23.5)	10.4 (7.1–15.0)	0.0007	8.0 (6.1–10.4)	10.2 (6.7–15.3)	5.9 (3.8–8.9)	0.1200
Belize	2008	7.7 (5.7–10.4)	11.7 (8.3–16.2)	4.4 (2.6–7.5)	0.0019	13.3 (10.5–16.7)	14.5 (10.7–19.5)	12.1 (9.8–15.0)	0.2018
Belize	P-Value	0.0011	0.0125	0.0086		0.0054	0.1440	0.0009	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Bolivia (La Paz)	2000	23.2 (20.2–26.4)	26.4 (23.6–29.4)	20.4 (16.7–24.7)	0.0024	9.8 (8.2–11.6)	11.7 (9.4–14.4)	8.1 (6.5–10.0)	0.0094
Bolivia (La Paz)	2003	16.3 (13.4–19.6)	20.3 (16.5–24.7)	12.0 (9.3–15.3)	0.0011	8.2 (7.0–9.7)	9.5 (7.8–11.5)	6.9 (5.7–8.3)	0.0040
Bolivia (La Paz)	P-Value	0.0021	0.0159	0.0010		0.1512	0.1520	0.2463	
Brazil (Curitiba)	2002	12.6 (9.5–16.6)	9.1 (5.6–14.5)	14.4 (10.5–19.4)	0.0627	3.4 (2.6–4.2)	3.9 (2.4–6.1)	2.9 (1.8–4.5)	0.4190
Brazil (Curitiba)	2006	13.7 (11.6–16.0)	10.0 (7.9–12.7)	16.6 (13.9–19.8)	0.0003	5.7 (4.7–6.8)	6.2 (4.6–8.4)	5.0 (3.7–6.8)	0.3752
Brazil (Curitiba)	P-Value	0.5927	0.6904	0.3803		0.0006	0.0578	0.0323	
Chile (Santiago)	2000	38.4 (34.5–42.3)	31.3 (26.3–36.8)	44.4 (41.3–47.5)	0.0001	6.6 (5.5–7.9)	6.9 (5.4–8.7)	5.4 (3.9–7.3)	0.2339
Chile (Santiago)	2003	33.9 (27.6–40.7)	27.6 (21.7–34.5)	39.2 (32.1–46.9)	0.0006	4.9 (3.6–6.6)	5.8 (3.7–8.9)	3.9 (2.8–5.4)	0.1355
Chile (Santiago)	2008	34.2 (31.3–37.3)	28.0 (24.3–32.0)	39.9 (36.0–43.9)	0.0000	9.2 (8.1–10.4)	9.5 (8.1–11.1)	8.8 (7.0–11.0)	0.6101
Chile (Santiago)	P-Value 00, 03	0.2184	0.3519	0.1791		0.0563	0.4488	0.1441	
Chile (Santiago)	P-Value 03, 08	0.9085	0.9091	0.8790		0.0000	0.0098	0.0001	
Colombia (Bogota)	2001	32.2 (29.0–35.6)	31.0 (27.5–34.8)	33.4 (29.5–37.5)	0.2161	5.1 (4.2–6.2)	5.9 (4.7–7.6)	4.3 (2.9–6.3)	0.1676
Colombia (Bogota)	2007	26.2 (22.5–30.3)	25.4 (21.0–30.3)	26.6 (20.9–33.1)	0.7546	5.0 (3.7–6.8)	6.7 (4.7–9.7)	3.6 (2.4–5.4)	0.0162
Colombia (Bogota)	P-Value	0.0193	0.0536	0.0614		0.8885	0.5673	0.5217	
Costa Rica	1999	17.8 (15.7–20.1)	18.2 (15.5–21.2)	17.3 (15.0–19.7)	0.5201	6.4 (5.6–7.4)	5.0 (4.0–6.2)	7.7 (6.7–8.9)	0.0003

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking: Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Current use of other tobacco products: Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Costa Rica	2002	16.4 (14.3–18.8)	15.7 (12.9–19.0)	16.8 (14.4–19.6)	0.5122	5.5 (4.8–6.4)	6.7 (5.3–8.4)	4.2 (3.3–5.5)	0.0238
Costa Rica	2008	9.6 (7.9–11.7)	9.4 (7.2–12.0)	9.7 (7.8–12.1)	0.7811	7.7 (6.6–8.9)	9.3 (8.1–10.8)	5.9 (4.5–7.7)	0.0011
Costa Rica	P-Value 99, 02	0.3852	0.2283	0.8046		0.1348	0.0639	0.0000	
Costa Rica	P-Value 02, 08	0.0000	0.0013	0.0001		0.0025	0.0105	0.0832	
Cuba (Havana)	2000	14.9 (11.3–19.3)	15.8 (11.8–20.9)	13.7 (9.9–18.7)	0.2861	6.1 (4.9–7.6)	4.4 (3.1–6.2)	7.9 (5.8–10.5)	0.0264
Cuba (Havana)	2004	10.0 (7.6–13.1)	11.2 (8.3–15.1)	8.8 (6.5–11.9)	0.0457	5.8 (3.3–10.0)	6.0 (3.0–11.8)	5.7 (3.3–9.7)	0.8463
Cuba (Havana)	P-Value	0.0392	0.0889	0.0499		0.8610	0.4292	0.2408	
Dominica	2000	11.6 (9.1–14.8)	13.4 (9.8–18.1)	9.6 (7.0–13.0)	0.0824	10.7 (8.5–13.3)	15.0 (11.0–20.1)	6.4 (4.7–8.6)	0.0014
Dominica	2004	11.5 (9.0–14.7)	11.8 (8.1–16.9)	9.6 (7.0–13.0)	0.4007	9.3 (7.6–11.4)	12.0 (9.0–15.9)	6.3 (4.5–8.9)	0.0129
Dominica	P-Value	0.9600	0.5794	0.9742		0.3850	0.2937	0.9703	
Ecuador (Quito)	2001	20.5 (17.1–24.5)	27.2 (23.0–31.8)	12.6 (9.0–17.4)	0.0006	10.0 (8.2–12.1)	12.1 (9.0–16.0)	7.1 (5.7–8.7)	0.0111
Ecuador (Quito)	2007	20.5 (15.6–26.6)	23.2 (19.4–27.6)	18.1 (11.1–28.0)	0.1508	15.3 (12.7–18.3)	15.9 (12.9–19.5)	14.6 (11.1–19.0)	0.5676
Ecuador (Quito)	P-Value	0.9968	0.1716	0.2185		0.0027	0.0961	0.0007	
Grenada	2000	8.3 (6.7–10.3)	9.9 (6.8–14.2)	6.5 (4.7–9.0)	0.1311	8.7 (7.0–10.7)	10.3 (8.0–13.2)	7.2 (5.7–9.2)	0.0274
Grenada	2004	10.2 (8.2–12.8)	10.9 (7.4–15.8)	9.5 (7.4–12.2)	0.5532	10.5 (8.5–12.9)	11.6 (9.0–14.8)	9.3 (7.1–12.1)	0.1301

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Grenada	P-Value	0.1833	0.7002	0.0613		0.1967	0.5017	0.1755	
Guatemala (Guatemala City)	2002	14.3 (11.7–17.5)	17.3 (12.8–22.8)	11.2 (8.5–14.5)	0.0431	5.6 (4.2–7.4)	6.6 (4.2–10.1)	3.3 (2.2–5.0)	0.0478
Guatemala (Guatemala City)	2008	14.8 (12.6–17.4)	16.6 (12.6–21.5)	12.8 (10.1–16.2)	0.2136	7.5 (6.2–8.9)	9.4 (7.5–11.8)	5.6 (4.4–7.1)	0.0019
Guatemala (Guatemala City)	P-Value	0.7794	0.8263	0.4167		0.0677	0.0975	0.0152	
Guyana	2000	8.2 (5.4–12.2)	11.3 (7.7–16.3)	6.4 (3.3–11.9)	0.0683	8.4 (6.3–11.0)	13.0 (8.0–20.3)	5.2 (3.4–8.0)	0.0281
Guyana	2004	8.1 (5.3–12.3)	11.0 (7.4–16.0)	5.4 (3.1–9.3)	0.0036	8.3 (6.4–10.7)	9.1 (6.3–12.9)	7.7 (4.9–11.9)	0.5744
Guyana	P-Value	0.9720	0.8972	0.6803		0.9656	0.2387	0.2028	
Haiti (Port-au-Prince)	2000	12.7 (9.0–17.6)	10.2 (6.6–15.4)	14.0 (9.0–21.0)	0.2742	10.7 (6.4–17.2)	6.3 (3.3–11.5)	13.3 (6.9–24.1)	0.1101
Haiti (Port-au-Prince)	2005	17.6 (13.6–22.6)	17.2 (12.4–23.5)	17.7 (13.3–23.0)	0.8341	10.4 (8.3–13.1)	9.0 (6.5–12.3)	11.1 (7.4–16.3)	0.4359
Haiti (Port-au-Prince)	P-Value	0.0897	0.0370	0.2941		0.9296	0.2254	0.6003	
Jamaica	2000	15.8 (12.6–19.7)	20.3 (15.0–26.9)	11.8 (9.5–14.5)	0.0073	7.8 (6.1–10.0)	10.5 (7.2–15.3)	5.4 (4.0–7.2)	0.0280
Jamaica	2006	15.4 (10.2–22.6)	20.6 (14.1–29.3)	10.9 (6.5–17.7)	0.0054	8.9 (6.5–11.9)	10.2 (7.1–14.5)	7.2 (4.8–10.8)	0.1492
Jamaica	P-Value	0.9076	0.9400	0.7523		0.5158	0.9036	0.2347	
Mexico (Mexico City)	2003	24.3 (19.7–29.6)	24.4 (19.8–29.7)	23.2 (16.5–31.5)	0.7355	7.6 (6.1–9.5)	9.9 (7.1–13.7)	4.9 (3.2–7.5)	0.0196
Mexico (Mexico City)	2006	27.1 (23.8–30.8)	26.3 (22.0–31.0)	27.1 (23.7–30.8)	0.6701	4.8 (3.6–6.4)	5.5 (3.3–9.0)	4.0 (3.0–5.3)	0.3077

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Mexico (Mexico City)	P-Value	0.3218	0.5588	0.3112		0.0096	0.0317	0.4186	
Panama	2002	13.2 (9.7–17.7)	14.7 (10.4–20.2)	11.1 (7.8–15.6)	0.0799	9.8 (8.4–11.5)	11.0 (8.3–14.6)	7.8 (6.0–10.1)	0.1398
Panama	2008	4.3 (3.0–6.2)	5.9 (4.0–8.5)	2.8 (1.7–4.6)	0.0023	5.8 (4.5–7.3)	7.1 (5.3–9.5)	4.5 (3.3–6.0)	0.0150
Panama	P-Value	0.0001	0.0014	0.0001		0.0002	0.0382	0.0073	
Paraguay (Asunción)	2003	19.1 (15.5–23.4)	19.2 (14.3–25.3)	18.5 (15.0–22.7)	0.7944	10.0 (8.0–12.4)	9.9 (7.1–13.5)	10.1 (7.9–12.8)	0.8876
Paraguay (Asunción)	2008	10.4 (8.3–13.1)	11.7 (9.3–14.5)	9.3 (6.6–12.8)	0.1121	10.1 (8.6–11.9)	11.8 (9.5–14.6)	8.6 (7.1–10.3)	0.0255
Paraguay (Asunción)	P-Value	0.0004	0.0122	0.0004		0.9009	0.3147	0.2629	
Peru (Lima)	2000	18.6 (15.2–22.5)	20.2 (15.8–25.4)	17.4 (13.3–22.4)	0.3384	6.3 (4.6–8.7)	7.7 (5.1–11.3)	5.1 (3.4–7.6)	0.1280
Peru (Lima)	2003	19.2 (15.1–24.0)	20.8 (15.3–27.7)	17.0 (13.1–21.7)	0.1376	7.9 (5.9–10.4)	8.2 (5.0–13.0)	7.1 (4.4–11.3)	0.7133
Peru (Lima)	2007	16.5 (13.0–20.7)	16.7 (12.8–21.6)	15.2 (11.0–20.7)	0.5331	3.4 (2.4–4.9)	3.8 (2.3–6.4)	3.1 (1.8–5.4)	0.5835
Peru (Lima)	P-Value 00, 03	0.8297	0.8712	0.9031		0.2870	0.8329	0.2695	
Peru (Lima)	P-Value 03, 07	0.3301	0.2536	0.5524		0.0008	0.0405	0.0277	
Saint Lucia	2000	9.5 (7.3–12.3)	11.5 (8.5–15.5)	7.9 (5.5–11.3)	0.0741	7.1 (5.2–9.5)	9.0 (6.2–12.7)	5.2 (3.3–7.9)	0.0428
Saint Lucia	2007	12.7 (10.4–15.3)	17.0 (12.2–23.1)	9.6 (7.4–12.4)	0.0258	10.2 (7.2–14.3)	13.0 (8.4–19.6)	8.4 (5.7–12.2)	0.0827

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Saint Lucia	P-Value	0.0680	0.0902	0.3800		0.1159	0.1979	0.0983	
Saint Vincent and The Grenadines	2000	13.8 (11.4–16.7)	17.2 (13.6–21.5)	10.7 (8.0–14.2)	0.0068	12.7 (10.0–15.8)	14.0 (10.4–18.5)	11.2 (8.3–14.8)	0.1964
Saint Vincent and The Grenadines	2007	12.0 (9.0–15.9)	14.8 (9.8–21.7)	9.5 (6.6–13.4)	0.0857	10.3 (8.2–13.0)	11.2 (8.2–15.2)	9.6 (7.1–12.9)	0.4405
Saint Vincent and The Grenadines	P-Value	0.4067	0.4931	0.5757		0.2151	0.2892	0.4744	
Suriname	2000	10.8 (8.0–14.5)	14.7 (9.4–22.3)	7.1 (4.5–11.0)	0.0396	6.0 (4.4–8.1)	7.3 (5.0–10.4)	4.4 (2.8–6.9)	0.05 08
Suriname	2004	6.9 (5.2–9.1)	9.3 (6.3–13.5)	4.7 (2.7–8.2)	0.0674	4.4 (3.3–6.0)	4.4 (2.7–7.1)	4.4 (3.2–6.2)	0.98 89
Suriname	P-Value	0.0345	0.1332	0.2318		0.1572	0.0843	0.9923	
Trinidad and Tobago	2000	12.0 (10.7–13.3)	16.0 (13.2–19.2)	7.6 (6.1–9.5)	0.0001	4.8 (3.8–5.9)	5.7 (4.1–7.8)	3.9 (3.0–4.9)	0.05 14
Trinidad and Tobago	2007	12.9 (9.9–16.7)	14.7 (10.9–19.6)	10.3 (6.9–15.1)	0.0563	8.9 (6.8–11.5)	8.9 (6.2–12.6)	8.7 (6.1–12.1)	0.88 28
Trinidad and Tobago	P-Value	0.5953	0.6421	0.2271		0.0017	0.0788	0.0026	
U.S. Virgin Islands††	2000	3.6 (2.5–5.1)	3.6 (2.2–5.9)	3.3 (2.0–5.3)	0.7841	6.7 (5.3–8.3)	8.6 (6.4–11.5)	4.9 (3.3–7.1)	0.0210
U.S. Virgin Islands††	2004	3.4 (2.5–4.6)	3.0 (2.0–4.6)	3.6 (2.4–5.5)	0.5727	6.1 (4.9–7.7)	8.8 (6.8 –11.4)	3.7 (2.3–5.7)	0.0005
U.S. Virgin Islands††	P-Value	0.8190	0.6016	0.7745		0.6084	0.9171	0.3409	
United States of America	2000	17.7 (16.3–19.3)	17.8 (16.0–19.7)	17.7 (16.1–19.4)	0.9111	14.5 (13.2–15.9)	19.9 (17.9–21.9)	9.1 (8.0–10.2)	0.0000
United States of America	2004	13.0 (11.6–14.6)	12.1 (10.5–13.9)	13.9 (12.3–15.8)	0.0273	10.6 (9.5–11.9)	14.0 (12.4–15.9)	7.4 (6.4–8.4)	0.0000

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking: Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Current use of other tobacco products: Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
United States of America	P-Value	0.0000	0.0000	0.0021		0.0000	0.0000	0.0265	
Uruguay	2000	25.6 (22.5–28.9)	21.2 (17.4–25.7)	28.6 (24.3–33.4)	0.0138	9.8 (8.1–11.8)	12.2 (9.1–16.2)	7.1 (5.4–9.2)	0.0177
Uruguay	2007	20.2 (18.0–22.6)	16.4 (13.5–19.8)	22.9 (20.1–26.0)	0.0028	7.9 (6.6–9.4)	10.3 (8.1–13.0)	6.1 (4.6–8.0)	0.0084
Uruguay	P-Value	0.0108	0.1218	0.0450		0.0702	0.5289	0.0580	
Venezuela (Lara State)	1999	9.4 (7.4–11.9)	9.3 (6.2–13.7)	9.5 (7.3–12.2)	0.9360	12.6 (10.2–15.4)	16.6 (12.7–21.4)	8.7 (6.1–12.4)	0.0055
Venezuela (Lara State)	2003	9.2 (6.7–12.6)	10.3 (6.2–16.7)	7.9 (5.6–11.2)	0.3444	8.7 (7.1–10.5)	11.5 (8.6–15.1)	6.0 (3.6–9.7)	0.0434
Venezuela (Lara State)	P-Value	0.9272	0.7318	0.3748		0.0130	0.0554	0.1813	
SOUTH-EAST ASIA REGION									
Bangladesh (Dhaka)	2004	1.8 (1.2–2.8)	2.3 (1.4–3.9)	0.0	0.0006	4.0 (3.1–5.2)	3.6 (2.5–5.0)	4.7 (3.4–6.4)	0.2353
Bangladesh (Dhaka)	2007	3.0 (1.6–5.4)	4.6 (2.8–7.5)	1.1 (0.4–2.9)	0.0054	6.5 (4.9–8.6)	7.3 (5.2–10.1)	4.7 (2.8–7.8)	0.1432
Bangladesh (Dhaka)	P-Value	0.2034	0.0712	0.0415		0.0144	0.0052	0.9739	
Bhutan	2004	10.1 (7.7–13.1)	14.6 (10.3–20.2)	4.9 (3.2–7.4)	0.0007	10.6 (7.6–14.7)	11.3 (7.1–17.5)	9.9 (7.0–13.8)	0.5346
Bhutan	2006	12.1 (9.6–15.2)	18.3 (13.8–23.8)	6.3 (4.1–9.6)	0.0004	14.2 (11.7–17.0)	19.7 (14.7–25.8)	9.1 (6.7–12.3)	0.0051
Bhutan	P-Value	0.2840	0.2710	0.3783		0.0940	0.0242	0.6897	
India	2004	4.2 (3.2–5.5)	5.7 (4.4–7.5)	1.8 (1.2–2.8)	0.0000	13.0 (12.1–13.9)	16.3 (15.0–17.6)	8.0 (6.6–9.6)	0.0000

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
India	2006	3.8 (3.1–4.7)	5.4 (4.3–6.7)	1.6 (1.0–2.6)	0.0000	11.9 (9.8–14.3)	14.3 (11.8–17.2)	8.5 (6.4–11.3)	0.0002
India	P-Value	0.5524	0.7170	0.7526		0.3548	0.1914	0.7124	
Indonesia (Jakarta)	2000	21.8 (17.9–26.2)	38.9 (32.2–46.2)	4.7 (3.1–7.1)	0.0000	2.5 (1.8–3.7)	4.0 (2.8–5.7)	0.9 (0.4–2.0)	0.0000
Indonesia (Jakarta)	2004	16.0 (13.1–19.3)	28.4 (23.2–34.3)	3.1 (1.5–6.4)	0.0000	5.0 (2.8–8.9)	5.4 (2.8–9.9)	4.6 (2.3–8.9)	0.5754
Indonesia (Jakarta)	P-Value	0.0238	0.0181	0.2785		0.0902	0.4267	0.0163	
Maldives	2004	6.9 (4.8–9.8)	9.0 (6.4–12.6)	3.1 (1.1–8.5)	0.0062	8.3 (5.5–12.3)	10.2 (6.5–15.9)	5.2 (2.5–10.5)	0.0871
Maldives	2007	3.8 (2.7–5.3)	6.6 (4.6–9.6)	0.9 (0.4–2.0)	0.0001	3.5 (2.2–5.5)	4.3 (2.5–7.4)	2.7 (1.6–4.7)	0.1543
Maldives	P-Value	0.0266	0.2274	0.1881		0.0103	0.0237	0.2169	
Sri Lanka	1999	4.0 (2.8–5.8)	6.2 (4.3–9.0)	1.6 (0.9–2.9)	0.0000	7.2 (6.1–8.4)	9.2 (7.2–11.6)	5.0 (3.9–6.4)	0.0029
Sri Lanka	2003	2.4 (1.5–3.7)	3.0 (1.8–4.9)	1.3 (0.6–2.9)	0.0394	7.0 (5.4–8.9)	7.9 (5.6–11.2)	5.8 (4.4–7.6)	0.0401
Sri Lanka	2007	1.2 (0.5–2.9)	1.6 (0.7–3.7)	0.9 (0.2–3.5)	0.3371	8.6 (6.4–11.5)	11.6 (8.0–16.6)	5.6 (3.5–8.7)	0.0196
Sri Lanka	P-Value 99, 03	0.0616	0.0198	0.7314		0.8327	0.4787	0.4637	
Sri Lanka	P-Value 03, 07	0.1392	0.1689	0.5528		0.2774	0.1394	0.8844	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking: Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Current use of other tobacco products: Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
WESTERN PACIFIC REGION									
China (Tianjin)	1999	5.7 (4.2–7.5)	10.7 (7.5–15.2)	1.5 (0.9–2.6)	0.0000	4.9 (3.9–6.2)	5.5 (4.1–7.4)	4.4 (3.2–6.0)	0.2386
China (Tianjin)	2005	3.2 (2.4–4.2)	5.6 (3.9–7.9)	0.6 (0.3–1.1)	0.0000	3.7 (2.7–5.1)	4.6 (3.3–6.4)	2.7 (1.6–4.8)	0.0585
China (Tianjin)	P-Value	0.0161	0.0227	0.0425		0.1518	0.4959	0.1066	
Fiji	1999	10.4 (7.3–14.6)	13.7 (9.5–19.4)	7.1 (4.2–11.7)	0.0238	7.9 (5.9–10.6)	10.5 (7.7–14.1)	5.4 (3.6–8.0)	0.0035
Fiji	2005	5.0 (2.9–8.5)	6.7 (3.8–11.6)	3.1 (1.6–6.0)	0.0037	7.7 (4.8–12.2)	6.7 (3.8–11.4)	7.6 (4.6–12.3)	0.5611
Fiji	P-Value	0.0160	0.0228	0.0561		0.9183	0.1148	0.3010	
Lao PDR (Vientiane Province)	2003	9.4 (6.1–14.2)	17.3 (10.9–26.4)	0.6 (0.3–1.2)	0.0003	5.9 (4.2–8.3)	8.1 (5.4–12.2)	3.3 (2.0–5.4)	0.0077
Lao PDR (Vientiane Province)	2007	6.7 (4.0–11.1)	12.0 (6.6–20.8)	0.4 (0.1–1.6)	0.0026	8.4 (6.4–11.0)	9.0 (5.8–13.7)	7.3 (4.6–11.4)	0.5135
Lao PDR (Vientiane Province)	P-Value	0.2887	0.2734	0.4014		0.0922	0.7183	0.0308	
Macau¶¶	2001	7.0 (4.7–10.5)	8.1 (4.7–13.7)	5.6 (3.1–10.0)	0.3325	1.2 (0.6–2.1)	2.0 (1.0–3.8)	0.4 (0.1–1.5)	0.0317
Macau¶¶	2005	10.4 (8.1–13.4)	11.0 (8.1–14.8)	9.8 (7.0–13.6)	0.5555	2.1 (1.4–3.0)	2.4 (1.5–3.9)	1.7 (0.9–3.1)	0.3543
Macau¶¶	P-Value	0.0785	0.2895	0.0688		0.0892	0.5963	0.0202	
Mongolia	2003	8.5 (6.4–11.2)	14.4 (10.9–18.7)	4.0 (2.7–5.7)	0.0000	8.0 (6.7–9.5)	9.5 (7.9–11.4)	6.8 (5.4–8.6)	0.0049
Mongolia	2007	6.9 (4.4–10.5)	11.0 (7.6–15.6)	3.3 (1.4–7.3)	0.0001	15.4 (7.3–29.5)	17.9 (10.3–29.3)	12.9 (4.6–31.6)	0.0432

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 3. (Continued)

WHO Region and Site	Year and Year Difference P-Value	Current cigarette smoking				Current use of other tobacco products			
		Total % (95% CI)*	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value	Total % (95% CI)	Boy % (95% CI)	Girl % (95% CI)	Boy vs Girl P-Value
Mongolia	P-Value	0.3690	0.2062	0.6491		0.1725	0.0794	0.3313	
Northern Mariana Islands**	2000	37.6 (34.4–41.0)	36.5 (32.2–41.1)	38.7 (34.2–43.3)	0.4948	35.2 (31.8–38.8)	44.6 (40.6–48.8)	26.2 (22.3–30.6)	0.0000
Northern Mariana Islands**	2004	29.1 (26.6–31.7)	26.6 (23.6–29.9)	31.5 (28.2–34.9)	0.0173	45.3 (42.3–48.4)	52.4 (48.7–56.0)	38.3 (34.6–42.2)	0.0000
Northern Mariana Islands**	P-Value	0.0001	0.0004	0.0132		0.0000	0.0060	0.0000	
Palau	2000	21.6 (18.2–25.4)	23.3 (18.3–29.2)	20.4 (16.0–24.8)	0.3548	NA	NA	NA	NA
Palau	2005	26.7 (23.3–30.2)	31.0 (26.9–35.5)	22.6 (18.1–27.8)	0.0088	NA	NA	NA	NA
Palau	P-Value	0.0451	0.0294	0.4479		NA	NA	NA	
Philippines	2000	18.2 (15.8–20.9)	26.2 (22.5–30.2)	12.4 (10.3–14.9)	0.0000	11.1 (9.9–12.3)	14.5 (12.8–16.5)	8.4 (7.1–10.0)	0.0000
Philippines	2004	11.5 (8.7–15.0)	16.6 (12.2–22.3)	7.8 (5.6–11.0)	0.0001	7.0 (5.9–8.3)	9.8 (7.9–12.1)	5.1 (3.9–6.6)	0.0001
Philippines	2007	17.5 (14.7–20.6)	23.4 (19.7–27.7)	12.0 (9.4–15.1)	0.0000	7.7 (6.2–9.5)	8.2 (6.1–10.9)	7.2 (5.6–9.2)	0.4476
Philippines	P-Value 00, 04	0.0011	0.0033	0.0115		0.0000	0.0009	0.0008	
Philippines	P-Value 04, 07	0.0059	0.0361	0.0361		0.4914	0.3137	0.0564	
Viet Nam (Hanoi)	2003	1.2 (0.8–2.0)	1.5 (0.8–3.0)	0.8 (0.3–2.1)	0.2718	1.2 (0.5–2.9)	1.9 (0.5–6.4)	0.5 (0.1–2.2)	0.2656
Viet Nam (Hanoi)	2007	3.0 (1.7–5.2)	5.0 (2.8–8.9)	1.0 (0.5–1.9)	0.0072	1.2 (0.8–1.9)	2.0 (1.3–3.1)	0.5 (0.1–2.1)	0.0148
Viet Nam (Hanoi)	P-Value	0.0394	0.0196	0.7422		0.9462	0.9053	0.9807	

* ( ) 95% confidence interval
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

There were changes in gender differences in five EMR sites. In Gaza Strip, Sudan, and Yemen the gender difference at Time 1 changed to no gender difference at Time 2; and in Egypt and Syria no gender difference at Time 1 changed to boys higher than girls at Time 2 (Table 3). Gender differences in other tobacco use did not change over time in the remaining 12 sites where boys had significantly higher use than girls in 10 sites; and there were no gender differences in Libya and Somalia.

European Region (EUR)

Current cigarette smoking

For the 18 EUR sites, current cigarette smoking decreased significantly for boys and girls in Georgia; among boys in the Russian Federation and Slovenia; and among girls in Serbia and Kyrgyzstan (Table 3). Smoking increased significantly for girls in Croatia. There were no changes over time in the rate of cigarette smoking in the remaining 12 sites.

Gender differences changed in four EUR sites. In Slovenia, no gender difference at Time 1 was reported, however, girls were higher than boys at Time 2. For Russian Federation, Slovakia, and Ukraine, boys were higher than girls at Time 1, but changed to no gender difference at Time 2 (Table 3). Gender differences in current cigarette smoking did not change over time in the remaining 14 sites where boys had significantly higher smoking than girls in 6 sites and girls had significantly higher smoking than boys in Hungary; there were no gender differences in 7 sites.

Current other tobacco use

For the 17 EUR sites with data, use of other tobacco products increased significantly among boys and girls in Bulgaria, Czech Republic, Estonia, Hungary, and Latvia; for boys in Croatia; and for girls in the Russian Federation and Ukraine (Table 3). Use of other tobacco products did not change over time in the remaining nine sites.

Changes in gender differences were reported in five EUR sites. In Bulgaria, there was no gender difference at Time 1, but changed to boys higher than girls at Time 2; and Montenegro, Russian Federation, Slovenia, and Ukraine had boys higher than girls at Time 1, but changed to no gender difference at Time 2 (Table 3). Gender differences in use of other tobacco products did not change over time in the remaining 12 sites where boys had significantly higher use than girls in 9 sites; and there were no gender differences in Croatia, Republic of Moldova, and Serbia.

Region of the Americas (AMR)

Current cigarette smoking

For the 30 AMR sites, current cigarette smoking decreased significantly for boys and girls in Belize, Bolivia, Costa Rica (2002 to 2008), Panama, Paraguay, and the United States; and for girls in Barbados (1999 to 2002), Cuba, and Uruguay (Table 3). Current cigarette smoking increased significantly for boys in Barbados (2002 to 2007) and in Haiti. For the remaining 20 sites the rate of cigarette smoking did not change over time.

Changes in gender differences were reported in 12 AMR sites. Cuba, Guyana, Panama, and St. Lucia had no gender differences at Time 1, but changed to boys higher than girls at Time 2; Barbados had no gender differences at Time 1 or Time 2, but changed to boys higher than girls at Time 3; Brazil and USA had no gender differences at Time 1, but changed to girls higher than boys at Time 2; and Ecuador, Guatemala, St. Vincent and The Grenadines, Suriname, and Trinidad and Tobago had gender differences at Time 1, but changed to no gender differences at Time 2 (Table 3). Gender differences in current cigarette smoking did not change over time in the remaining 18 sites: boys had significantly higher smoking than girls in Belize, Bolivia, and Jamaica; girls had significantly higher smoking than boys in Argentina, Chile, and Uruguay; and there were no gender differences in 12 sites.

Current other tobacco use

For the 30 AMR sites, use of other tobacco products decreased significantly for boys and girls in Panama, Peru (2003 to 2007), and the USA; for boys in the Bahamas, Mexico, and Venezuela; and for girls in Costa Rica (1999 to 2008) (Table 3). Other tobacco use increased significantly for boys and girls in Barbados (2002 to 2007) and Chile (2003 to 2008); for boys in Costa Rica (2002 to 2008); and for girls in Argentina (2003 to 2007), Belize, Brazil, Ecuador, Guatemala, and Trinidad & Tobago. There were no changes in the rate of other tobacco use across time for the remaining 16 sites.

Gender differences changed in 12 sites. In seven sites boys had higher use than girls at Time 1 but no gender difference at Time 2; Colombia, Panama, and Paraguay

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

had no gender difference at Time 1, but changed to boys higher than girls at Time 2; Barbados had no gender differences at Time 1 or Time 2, but changed to boys higher than girls at Time 3; and Cuba had girls higher than boys at Time 1, but changed to no gender difference at Time 2 (Table 3). Gender differences in other tobacco use did not change over time in the remaining 18 sites. Boys had significantly higher use than girls in nine sites; and there were no gender differences in nine sites.

South-East Asia Region (SEAR)

Current cigarette smoking

For the 6 SEAR sites, current cigarette smoking decreased significantly for girls in Indonesia and Sri Lanka (1999 to 2003); increased for boys in Bangladesh; but did not change significantly in the other 3 sites (Table 3). For the 6 sites, gender differences in current cigarette smoking did not change over time in 5 sites with boys having significantly higher smoking than girls in Bangladesh, Bhutan, India, Indonesia, and Maldives (Table 3). Gender differences changed in Sri Lanka where boys had higher rates than girls at Time 1 and Time 2, but changed to no gender difference at Time 3.

Current other tobacco use

For the 6 SEAR sites, use of other tobacco products decreased significantly for boys in Maldives; increased significantly for boys in Bangladesh and Bhutan; and increased significantly for girls in Indonesia (Table 3). Use of other tobacco products did not change over time in India and Sri Lanka.

Bhutan had no gender difference at Time 1, but boys were higher than girls at Time 2; and in Indonesia the gender difference at Time 1 changed to no gender difference at Time 2 (Table 3). Gender differences in use of other tobacco products did not change over time in the remaining 4 sites. Boys had significantly higher smoking than girls in India and Sri Lanka and there were no gender differences in Bangladesh and Maldives.

Western Pacific Region (WPR)

Current cigarette smoking

For the 9 WPR sites, current cigarette smoking decreased significantly for boys and girls in China, the Philippines (2000 to 2004) and the Northern Mariana Islands; and significantly decreased for boys in Fiji (Table 3). Current smoking increased significantly for boys and girls in the Philippines (2004 to 2007) and for boys in Viet Nam and Palau. There was no change over time in the rate of cigarette smoking in Laos, Macau, or Mongolia.

Gender differences changed in Palau and Viet Nam with no gender difference at Time 1 changing to boys higher than girls at Time 2; and in the Northern Mariana Islands with no gender difference at Time 1 changing to girls higher than boys at Time 2 (Table 3). Gender differences in current cigarette smoking did not change over time in the remaining 6 sites where boys had significantly higher smoking than girls in China, Fiji, Laos, Mongolia, and Philippines; and there were no gender differences in Macau.

Current other tobacco use

For the 8 WPR sites with data, use of other tobacco products decreased significantly for boys and girls in the Philippines (2000 to 2004); increased for boys and girls in Northern Mariana Islands; and increased for girls in Laos and Macau (Table 3). There were no changes over time in other tobacco use in China, Fiji, Mongolia, and Viet Nam.

There were changes in gender differences in five WPR sites. In Viet Nam there was no gender difference at Time 1, but boys were higher than girls at Time 2; Fiji, Laos, and Macau had boys higher than girls at Time 1, but changed to no gender difference at Time 2; Philippines had boys higher than girls at Time 1 and Time 2, but changed to no gender difference at Time 3 (Table 3). Gender differences in other tobacco use did not change over time in the remaining 3 sites: boys had significantly higher use than girls in Mongolia and Northern Mariana Islands; and there were no gender differences in China.

Discussion

This article examines change in tobacco use (cigarette smoking and other tobacco use) among youth aged 13–15 years in 100 survey sites. In the next sections we look at specifics of the change and the factors that may have contributed to the change.

Change in cigarette smoking

We found significant change in cigarette smoking rates in 39 sites: a decrease in 27 sites; an increase in

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 4. Summary of change in tobacco use by World Health Organization (WHO) region, WHO member state, geographic region, territory, special administrative region, or commonwealth, type of tobacco used and if the change was a decrease or increase–GYTS 1999–2008

	Current Cigarette Smoking Time 1 vs Time 2/3			*Current Other Tobacco Use Time 1 vs Time 2/3*		
WHO Region and Site	*No Change*	*Decrease*	*Increase*	*No Change*	*Decrease*	*Increase*
AFRICAN REGION						
Botswana			B & G			B & G
Burkina Faso (Ouagadougou)		B & G		NC		
Ghana	NC				G	
Kenya	NC			NC		
Lesotho	NC					B
Malawi (Blantyre)	NC				B & G	
Mali (Bamako)		B & G		NC		
Mauritania			G	NC		
Mauritius	NC			NA		
Mozambique (Maputo City)	NC					B
Niger		B & G		NC		
Nigeria (Cross River State)	NC			NC		
Senegal		B				B
Seychelles	NC			NC		
South Africa	NC			NC		
Swaziland	NC					G
Togo		G		NC		
Uganda (Kampala)	NC			NC		
Zambia (Lusaka)	NC			NC		
Zimbabwe (Harare)	NC ('03-'08)	B & G ('99-'03)		NC ('99-'03 and '03-'08)		
EASTERN MEDITERRANEAN REGION						
Egypt	NC				B & G	
Gaza Strip§§	NC					G
Iran	NC					B & G
Jordan	NC ('99-'03)	B & G ('03-'07)				B & G ('99-'03) ('03-'07)
Kuwait	NC			NC		
Lebanon	NC			NC		
Libya	NC			NC		
Morocco	NC			NC		
Oman		B			B & G	
Qatar	NC			NC		

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B–Boys
G–Girls
NC–No change
NA–Data not available

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 4. (Continued)

WHO Region and Site	*Current Cigarette Smoking Time 1 vs Time 2/3*			*Current Other Tobacco Use Time 1 vs Time 2/3*		
	No Change	*Decrease*	*Increase*	*No Change*	*Decrease*	*Increase*
Somalia (Somaliland)		G		NC		
Sudan	NC				B	
Syria			B & G			B
Tunisia	NC					B & G
United Arab Emirates			G			B & G
West Bank§§			G	NC		
Yemen	NC			NC		
EUROPEAN REGION						
Bosnia and Herzegovina	NC			NC		
Bulgaria	NC					B & G
Croatia			G			B
Czech Republic	NC					B & G
Estonia	NC					B & G
Georgia		B & G		NA		
Hungary	NC					B & G
Kyrgyzstan		G		NC		
Latvia	NC					B & G
Lithuania	NC			NC		
Montenegro	NC			NC		
Poland (Urban)	NC			NC		
Republic of Moldova	NC			NC		
Russian Federation (Moscow)		B				G
Serbia		G		NC		
Slovakia	NC			NC		
Slovenia		B		NC		
Ukraine (Kiev)	NC					G
REGION OF THE AMERICAS						
Antigua and Barbuda	NC			NC		
Argentina (Capital Federal)	NC ('00-'03 and '03-'07)			NC ('00-'03 and B '03-'07)		G ('03-'07)
Bahamas	NC				B	
Barbados	NC (B '99-'02 and G '02-'07)	G ('99-'02)	B ('02-'07)	NC ('99-'02)		B & G ('02-'07)

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B–Boys
G–Girls
NC–No change
NA–Data not available

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 4. (Continued)

	Current Cigarette Smoking Time 1 vs Time 2/3			*Current Other Tobacco Use Time 1 vs Time 2/3*		
WHO Region and Site	*No Change*	*Decrease*	*Increase*	*No Change*	*Decrease*	*Increase*
Belize		B & G				G
Bolivia (La Paz)		B & G		NC		
Brazil (Curitiba)	NC					G
Chile (Santiago)	NC ('00-'03 and '03-'08)			NC ('00-'03)		B & G ('03-'08)
Colombia (Bogota)	NC			NC		
Costa Rica	NC ('99-'02)	B & G ('02-'08)		NC (B '99-'02 and G '02-'08)	G ('99-'02)	B ('02-'08)
Cuba (Havana)		G		NC		
Dominica	NC			NC		
Ecuador (Quito)	NC					G
Grenada	NC			NC		
Guatemala (Guatemala City)	NC					G
Guyana	NC			NC		
Haiti (Port-au-Prince)			B	NC		
Jamaica	NC			NC		
Mexico (Mexico City)	NC				B	
Panama		B & G			B & G	
Paraguay (Asunción)		B & G		NC		
Peru (Lima)	NC ('00-'03 and '03-'07)			NC ('00-'03)	B & G ('03-'07)	
St Lucia	NC			NC		
St Vincent and The Grenadines						
Suriname	NC			NC		
Trinidad and Tobago	NC					G
US Virgin Islands††	NC			NC		
United States of America		B & G			B & G	
Uruguay		G		NC		
Venezuela (Lara State)	NC			NC		
SOUTH-EAST ASIA REGION						
Bangladesh (Dhaka)			G			B
Bhutan	NC					B
India	NC			NC		

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B–Boys
G–Girls
NC–No change
NA–Data not available

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 4. (Continued)

	Current Cigarette Smoking Time 1 vs Time 2/3			*Current Other Tobacco Use Time 1 vs Time 2/3*		
WHO Region and Site	*No Change*	*Decrease*	*Increase*	*No Change*	*Decrease*	*Increase*
Indonesia (Jakarta)		B				G
Maldives	NC				B	
Sri Lanka	NC (G '99-'03 and '03-'07)	B ('99-'03)		NC ('99-'03 and '03-'07)		
WESTERN PACIFIC REGION						
China (Tianjin)		B & G		NC		
Fiji		B		NC		
Lao PDR (Vientiane Province)	NC					G
Macau¶¶	NC					G
Mongolia	NC			NC		
Northern Mariana Islands**		B & G				B & G
Palau			B	NA		
Philippines		B & G ('00-'04)	B & G ('04-'07)	NC ('04-'07)	B & G ('00-'04)	
Viet Nam (Hanoi)			B	NC		

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B–Boys
G–Girls
NC–No change
NA–Data not available

10 sites; a decrease followed by an increase in the Philippines; and no change in Costa Rica followed by a decrease for boys and girls (Table 4 summarizes the change). There were no changes over time in 61 sites. In a recent WHO report (referred to as the MPOWER Report for the remainder of this article) six important and effective tobacco control policies were identified which have been shown to reduce tobacco use: protect people from tobacco smoke (enact and enforce completely smoke-free environments); offer help to quit tobacco use (strengthen health systems so they can make tobacco cessation advice available as part of primary health care); warn about the dangers of tobacco (require effective package warning labels, implement counter-tobacco advertising, obtain free media coverage of anti-tobacco activities); enforce bans on tobacco advertising, promotion, and sponsorship; raise taxes on tobacco products; and monitor tobacco use (2). These six tobacco control measures relate to specific articles included in the WHO FCTC. Countries that have ratified the WHO FCTC are encouraged to implement the six policies identified in the MPOWER Report.

Of the 29 countries that showed a decrease at either Time 2 or 3, 6 had not ratified the WHO FCTC (Cuba, Indonesia, Northern Mariana Islands, Somalia, USA, and Zimbabwe); 6 others had only subnational data (Bolivia, Burkina Faso, China, Mali, Paraguay, and Russian Federation); and Costa Rica ratified in 2008, after their most recent GYTS. Of the remaining 16, 10 had both national objectives and a national agency (Belize, Jordan, Oman, Panama, Philippines, Senegal, Slovenia, Sri Lanka, Togo, and Uruguay). The following section reviews

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

the tobacco control program efforts in these 10 countries relative to the 6 policies identified in the MPOWER Report.

Taxation

Tax on tobacco products has been shown to be the most effective way to reduce tobacco use (2). Studies have shown that increasing taxes will have a positive effect on lowering tobacco use among young people and low-wage earners. Increasing tobacco taxes by 10% generally decreases tobacco consumption by 4% in high income countries and 8% in low-middle income countries (8). Data in the MPOWER Report show tobacco taxes are at low levels in most countries – less than one-fourth of low to middle income countries have a tax on tobacco at more than 50% of the retail price (2).

Among the 10 countries that had a decrease in cigarette smoking, only 4 countries had a tobacco tax at 50% or higher than the retail price: Uruguay (70%), Slovenia (58%), Sri Lanka (54%), and Oman (50%). The tax in the Philippines was 41% and 39% in Jordan. In Panama, Senegal and Togo taxes were less than 25% of the retail price. There was no information on tobacco taxes in Belize in the MPOWER Report but Belize has increased the tobacco tax twice since 2005 (9). In March 2005 the excise tax on cigarettes was increased from BZ$4.00 to BZ$12.00 per carton of 200 sticks then increased further in July 2008 to BZ$18.00. The MPOWER Report concludes that all countries need to increase their taxes on tobacco (2).

Second hand smoke (SHS) bans

According to the MPOWER Report, smoke free laws are only effective if they encompass eight specific public places (i.e. health care facilities, education facilities, university facilities, government facilities, indoor offices, restaurants, pubs and bars, and other indoor workplaces) (2). Uruguay is the only one of the 10 sites with a complete smoke-free policy with bans on smoking in all enclosed public places, including bars, restaurants, and casinos. The Ministry of Public Health in Uruguay developed a National Plan of Action with two main objectives: (i) to decrease smoking prevalence in the general population, and (ii) to protect people from exposure to SHS (10). Slovenia has the second strongest bans on SHS. In 2007, Slovenia enacted a new law (Restriction on the Use of Tobacco Products Act) which included bans on smoking in all enclosed public places and work places (11). Smoking rooms are not allowed in health care, child care or education facilities. However, Slovenia does not have a complete SHS ban, as smoking is allowed in certain rooms and areas specifically identified in the MPOWER Report (2).

Smoke-free policies are in place for Panama and Sri Lanka but these policies do not include all places identified in the MPOWER report. Panama bans smoking in health centers, education and government facilities (12). Sri Lanka bans smoking in health centers, education, university and government facilities, and other indoor facilities (2). The smoke free policies in Jordan, Oman, Slovenia, and the Philippines are minimal (2). All four sites ban smoking in health centers and education facilities. In addition, Oman and the Philippines ban smoking in university and government facilities; whereas, Jordan also bans smoking in universities and other indoor facilities. For the remaining countries, the smokefree laws are either very weak (Senegal and Togo), or non-existent (Belize). WHO states that the only proven way to protect people adequately from the harmful effects of SHS is to have complete smoke-free environments (2).

Tobacco advertising, promotion, and sponsorship bans

The MPOWER Report states that, 'a ban on marketing and promotion is a powerful weapon against the tobacco epidemic' (2). To be effective, marketing bans must be complete and apply to seven advertising categories: national and international bans on TV/radio or newspaper/magazine bans, and bans on billboards, point of sale and Internet advertising. Additionally, six categories of sponsorship advertising are included in the MPOWER Report: free distribution, promotional discounts, non-tobacco products identified with tobacco brand names, brand name of non-tobacco products used for tobacco products, tobacco products on TV or in films, and sponsored events (2).

In Uruguay recent legislation (Law 18.256) bans ads on national and international TV and radio, local and international magazines and newspapers, and on billboards and outdoor advertising (10). Ads

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

are allowed on the internet and at points of sale but only if accompanied with a health warning message from the Ministry of Public Health. In addition, Uruguay banned free distribution, promotional discounts, non-tobacco products identified with tobacco brand names, brand names of non-tobacco products used for tobacco products and sponsored events. Sri Lanka has 11 of 13 categories of bans, followed by Jordan with 10 bans, and the Philippines with eight bans. Slovenia has legislation banning only ads on TV and in magazines. Senegal, Togo, Belize, and Oman have no bans. To be effective, advertising bans must apply to all marketing, promotion and sponsorship categories. Seven of the 10 sites have a great deal of work to do to reach having a comprehensive policy.

Warning labels

The MPOWER Report states, 'Comprehensive warnings about the dangers of tobacco … encourage tobacco users to quit and young people not to start, and help gain public acceptance of other tobacco control measures such as establishing smoke free environments' (2). Uruguay and Panama require health warnings on at least 50% of all cigarette packages. In Law No 13, Panama strengthened the tobacco control measures by banning pro-tobacco statements on all cigarette packages (12). Jordan and Slovenia require health warnings on 30% of the cigarette packages; however, Belize, Oman, Philippines, Senegal, Sri Lanka, and Togo have no legislation requiring warning labels. Clearly, much work is needed in eight of the 10 sites regarding enacting strong legislation requiring health warnings on 50% or more of cigarette packages.

Cessation

The MPOWER Report states, 'Countries must establish programmes providing low-cost, effective treatment for tobacco users who want to escape their addiction' (2). Smoking cessation counseling is widely available in Uruguay and moderately available in Senegal, Togo, Panama, Jordan, and Oman. Cessation counseling is only minimally available in Belize, Slovenia, Sri Lanka, and the Philippines. For the six countries with cessation counseling, it is unclear how many of these programs are open to adolescents and if so, how many adolescents participate. The problem facing these countries, as with other countries, is summarized in the report, Youth Tobacco Cessation: A Guide for Making Informed Decisions, ' … a literature review of 66 published studies on youth tobacco-use cessation and reduction … concluded that most of the studies lacked the quality and consistency of findings to allow conclusive recommendations about effective practices … ' (13). More research is needed to evaluate and identify effective youth tobacco cessation programs.

Other tobacco control actions

A few of the 10 sites had additional tobacco control actions. In Belize, the National Tobacco Control Plan includes school education programs and anti-tobacco school policies (9). In addition, in Belize anti-tobacco promotional materials have been produced and disseminated in primary and secondary schools. In Panama, Law No 13 required the integration of content on the health consequences of tobacco consumption into the curricula of general education and basic secondary education (12). In Jordan, Law 34 imposed sanctions on the sale of tobacco to adolescents (less than age 18) and Law No 37 ('Monitoring Juveniles Behavior') was enacted in 2003(14). In addition, the Ministry of Health Jordan and UNICEF prepared a program on the hazards of smoking that was presented at schools and youth centers and outreach programs for parents about the harmful effects of smoking were also instituted. In Slovenia, an anti-tobacco education program was established for the general population in a 2003 Directive (11).

Summary

Jordan, Panama, Slovenia, Sri Lanka, and Uruguay have enacted tobacco control policies that likely have had an impact on reducing adolescent cigarette smoking. The Philippines enacted a strong tobacco control law in 2003, which likely contributed to the decrease in adolescent smoking between 2000 and 2004 (15). However, after a government change in 2004, the political will supporting tobacco control diminished and may have led to the reversal of trends such that by 2007 adolescent smoking was back to the 2000 level. Belize has raised taxes in recent years and has established some school programs; but legislation banning pro-tobacco advertising and requiring

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 5. Summary of change in gender differences in tobacco use by World Health Organization (WHO) region, WHO member state, geographic region, territory, special administrative region, or commonwealth–GYTS 1999–2008

	Current cigarette smoking			*Current Other Tobacco Use*		
		Change in gender difference			*Change in gender difference*	
WHO Region and Site	*No change in gender difference*	*No Difference to Difference*	*Difference to No Difference*	*No change in gender difference*	*No Difference to Difference*	*Difference to No Difference*
AFRICAN REGION						
Botswana	B>G			ND		
Burkina Faso (Ouagadougou)	B>G				X (B>G)	
Ghana	ND			ND		
Kenya		X (B>G)		ND		
Lesotho			X	ND		
Malawi (Blantyre)	ND			ND		
Mali (Bamako)	B>G			B>G		
Mauritania			X	ND		
Mauritius	B>G			NA		
Mozambique (Maputo City)	B>G			ND		
Niger	B>G			ND		
Nigeria (Cross River State)	B>G					X
Senegal	B>G					X
Seychelles	ND					X
South Africa		X (B>G)				X
Swaziland	B>G			B>G		
Togo	B>G				X (B>G)	
Uganda (Kampala)	ND			ND		
Zambia (Lusaka)	ND			ND		
Zimbabwe (Harare)		X (B>G '99-'03)	X ('03-'08)	ND		
EASTERN MEDITERRANEAN REGION						
Egypt		X (B>G)			X (B>G)	
Gaza Strip§§	B>G					X
Iran	B>G			B>G		
Jordan	B>G			B>G		
Kuwait	B>G			B>G		
Lebanon	B>G			B>G		
Libya	B>G			ND		
Morocco			X	B>G		
Oman	B>G			B>G		
Qatar	B>G			B>G		
Somalia (Somaliland)	ND			ND		

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B>G: boys significantly higher than girls
G>B: girls significantly higher than boys
ND: no significant gender difference
NA: data not available
X: denotes change overtime occurred

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 5. (Continued)

	Current cigarette smoking			*Current Other Tobacco Use*		
		Change in gender difference			*Change in gender difference*	
WHO Region and Site	*No change in gender difference*	*No Difference to Difference*	*Difference to No Difference*	*No hange in gender difference*	*No Difference to Difference*	*Difference to No Difference*
Sudan	B>G					X
Syria	B>G				X (B>G)	
Tunisia	B>G			B>G		
United Arab Emirates	B>G			B>G		
West Bank§§	B>G			B>G		
Yemen			X			X
EUROPEAN REGION						
Bosnia and Herzegovina	B>G			B>G		
Bulgaria	G>B				X (B>G)	
Croatia	ND			ND		
Czech Republic	ND			B>G		
Estonia	ND			B>G		
Georgia	B>G			NA		
Hungary	ND			B>G		
Kyrgyzstan	B>G			B>G		
Latvia	B>G			B>G		
Lithuania	B>G			B>G		
Montenegro	ND					X
Poland (Urban)	B>G			B>G		
Republic of Moldova	ND			ND		
Russian Federation (Moscow)			X			X
Serbia	ND			ND		
Slovakia			X	B>G		
Slovenia		X (G>B)				X
Ukraine (Kiev)			X			X
REGION OF THE AMERICAS						
Antigua and Barbuda	ND			ND		
Argentina (Capital Federal)	G>B			B>G		
Bahamas	ND					X
Barbados		X (B>G '02-'07)			X (B>G '02-'07)	
Belize	B>G			ND		
Bolivia (La Paz)	B>G			B>G		
Brazil (Curitiba)		X (G>B)		ND		
Chile (Santiago)	G>B			ND		

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B>G: boys significantly higher than girls
G>B: girls significantly higher than boys
ND: no significant gender difference
NA: data not available
X: denotes change overtime occurred

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 5. (Continued)

	Current cigarette smoking			*Current Other Tobacco Use*		
		Change in gender difference			*Change in gender difference*	
WHO Region and Site	*No change in gender difference*	*No Difference to Difference*	*Difference to No Difference*	*No change in gender difference*	*No Difference to Difference*	*Difference to No Difference*
Colombia (Bogota)	ND				X (B>G)	
Costa Rica	ND			B>G		
Cuba (Havana)		X (B>G)				X
Dominica	ND			B>G		
Ecuador (Quito)			X			X
Grenada	ND					X
Guatemala (Guatemala City)			X	B>G		
Guyana		X (B>G)				X
Qatar	B>G			B>G		
Somalia (Somaliland)	ND			ND		
Sudan	B>G					X
Syria	B>G				X (B>G)	
Tunisia	B>G			B>G		
United Arab Emirates	B>G			B>G		
West Bank§§	B>G			B>G		
Yemen			X			X
Haiti (Port-au-Prince)	ND			ND		
Jamaica	B>G					X
Mexico (Mexico City)	ND					X
Panama		X (B>G)			X (B>G)	
Paraguay (Asuncion)	ND				X (B>G)	
Peru (Lima)	ND			ND		
St Lucia		X (B>G)				X
St Vincent and The Grenadines			X	ND		
Suriname			X	ND		
Trinidad and Tobago			X	ND		
US Virgin Islands††	ND			B>G		
United States of America		X (G>B)		B>G		
Uruguay	G>B			B>G		
Venezuela (Lara State)	ND			B>G		
SOUTH-EAST ASIA REGION						
Bangladesh (Dhaka)	B>G			ND		

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B>G: boys significantly higher than girls
G>B: girls significantly higher than boys
ND: no significant gender difference
NA: data not available
X: denotes change overtime occurred

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 5. (Continued)

	Current cigarette smoking			*Current Other Tobacco Use*		
		Change in gender difference			*Change in gender difference*	
WHO Region and Site	*No change in gender difference*	*No Difference to Difference*	*Difference to No Difference*	*No change in gender difference*	*No Difference to Difference*	*Difference to No Difference*
Bhutan	B>G				X (B>G)	
India	B>G			B>G		
Indonesia (Jakarta)	B>G					X
Maldives	B>G			ND		
Sri Lanka			X ('03-'07)	B>G		
WESTERN PACIFIC REGION						
China (Tianjin)	B>G			ND		
Fiji	B>G					X
Lao PDR (Vientiane Province)	B>G					X
Macau¶¶	ND					X
Mongolia	B>G			B>G		
Northern Mariana Islands**		X (G>B)		B>G		
Palau		X (B>G)		NA		
Philippines	B>G					X ('04-'07)
Viet Nam (Hanoi)		X (B>G)			X (B>G)	

§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
B>G: boys significantly higher than girls
G>B: girls significantly higher than boys
ND: no significant gender difference
NA: data not available
X: denotes change overtime occurred

smoke free environments does not exist. Oman has minimal smokefree legislation but no other tobacco control policy. There is no evidence in the MPOWER Report that supports a decrease in adolescent smoking in Senegal or Togo. Further study of the tobacco control program and policy efforts in all 10 countries is needed, specifically in Belize, Oman, Senegal and Togo.

Change in other tobacco use

For the 97 sites that had data on other tobacco use, we found significant change in 47 sites: a decrease in 13 sites, an increase in 33 sites, and in Costa Rica a decrease for girls but an increase for boys (Table 4 summarizes the change). Of the 34 sites that reported a significant increase in other tobacco use: 13 showed a significant increase for boys and girls; 8 sites for boys only; and 13 sites for girls only.

Several factors have contributed to the increase in other tobacco use among adolescents. Recent studies have noted that for adolescents and young adults smoking tobacco in waterpipes (also known as shisha, hookahs, hubble-bubbles, narghiles, shishas, gozas, or narkeelas) may be responsible for much of this increase, especially in the Middle East, Europe and parts of North and South America (16). The recent popularity of the waterpipe has been attributed to target marketing by the tobacco industry (16). The tobacco industry has promoted the use of waterpipes by including sweet additives and more flavors with the tobacco 'rocks' used in the waterpipes in a special

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 6 Gender difference in current cigarette smoking for adult males and females and adolescent boys and girls, by World Health Organization (WHO) region, WHO member state, geographic region, territory, special administrative region, or commonwealth and sex–MPOWER data for adults, GYTS data for youth

WHO Region and Site	*Boy % (95% CI)*	*Male % (95% CI)*	*Girl % (95% CI)*	*Female % (95% CI)*
AFRICAN REGION				
Botswana	18.1 (13.4–23.9)	NA	10.9 (7.8–15.0)	NA
Burkina Faso (Ouagadougou)	14.1 (10.4–18.7)	14.2 (12.6–15.7)	2.4 (1.3–4.3)	0.8 (0.5–1.2)
Ghana	2.8 (1.7–4.7)	7.1 (5.8–8.3)	2.3 (1.4–3.5)	0.5 (0.2–0.8)
Kenya	11.2 (8.9–14.0)	23.9 (21.0–26.8)	5.2 (3.5–7.6)	1.1 (0.6–1.6)
Lesotho	11.8 (7.0–19.3)	NA	7.5 (4.9–11.2)	NA
Malawi (Blantyre)	2.0 (1.0–4.1)	19.2 (16.9–21.6)	1.4 (0.3–5.7)	2.3 (1.6–3.1)
Mali (Bamako)	18.6 (12.1–27.5)	14.0 (12.3–15.6)	1.7 (0.8–3.9)	0.7 (0.3–1.2)
Mauritania	20.3 (17.5–23.4)	16.3 (14.2–18.3)	18.3 (13.4–24.5)	0.8 (0.4–1.1)
Mauritius	20.3 (13.9–28.6)	35.7 (31.9–39.5)	7.7 (4.1–14.0)	1.1 (0.6–1.6)
Mozambique (Maputo City)	4.5 (2.6–7.9)	20.0 (18.0–22.1)	1.2 (0.4–3.5)	1.6 (1.1–2.1)
Niger	11.7 (7.6–17.4)	NA	1.1 (0.3–3.9)	NA
Nigeria (Cross River State)	6.8 (2.4–17.7)	9.1 (7.6–10.5)	1.2 (0.2–6.4)	0.2 (0.0–0.4)
Senegal	12.1 (7.6–18.9)	14.4 (12.3–16.5)	2.7 (1.3–5.4)	0.6 (0.2–1.0)
Seychelles	23.2 (17.4–30.2)	30.8 (26.1–35.4)	20.0 (15.0–26.2)	3.0 (1.4–4.5)
South Africa	21.0 (16.7–26.1)	25.0 (21.7–28.3)	10.6 (8.3–13.4)	7.8 (6.1–9.4)
Swaziland	8.9 (7.8–10.2)	13.3 (10.8–15.8)	3.2 (2.5–4.2)	2.8 (1.8–3.7)
Togo	9.1 (5.1–15.6)	NA	1.7 (1.1–2.6)	NA
Uganda (Kampala)	5.7 (3.5–9.2)	18.4 (16.0–20.8)	4.2 (2.7–6.4)	1.5 (1.0–1.9)

* Source: 2008 MPOWER Report
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
NA–Data not available

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 6. (Continued)

WHO Region and Site	*Boy % (95% CI)*	*Male % (95% CI)*	*Girl % (95% CI)*	*Female % (95% CI)*
Zambia (Lusaka)	6.7 (4.0–11.1)	18.0 (15.6–20.4)	6.8 (4.0–11.3)	2.1 (1.3–2.9)
Zimbabwe (Harare)	4.8 (2.6–9.0)	21.2 (18.3–24.0)	1.5 (0.5–4.6)	2.0 (1.2–2.7)
EASTERN MEDITERRANEAN REGION				
Egypt	5.9 (4.4–7.9)	24.5 (22.4–26.6)	1.4 (0.9–2.3)	0.9 (0.5–1.2)
Gaza Strip§§	9.7 (6.3–14.6)	NA	3.0 (1.6–5.4)	NA
Iran	5.1 (2.8–9.1)	24.0 (19.5–28.4)	0.9 (0.4–1.9)	1.9 (1.3–2.5)
Jordan	13.2 (9.9–17.5)	61.9 (52.9–70.9)	7.1 (4.9–10.3)	9.8 (4.0–15.6)
Kuwait	17.7 (14.2–21.7)	NA	4.5 (3.0–6.9)	NA
Lebanon	11.8 (8.5–16.3)	29.1 (24.2–34.0)	5.6 (4.2–7.5)	7.0 (2.7–11.2)
Libya	7.7 (4.9–11.9)	NA	0.9 (0.3–2.5)	NA
Morocco	4.3 (2.9–6.4)	26.1 (24.0–28.3)	2.1 (1.1–3.9)	0.2 (0.0–0.4)
Oman	3.5 (1.8–6.6)	24.1 (20.4–27.8)	1.2 (0.3–4.1)	0.3 (0.1–0.6)
Qatar	13.4 (9.5–18.7)	NA	2.3 (1.0–5.1)	NA
Somalia (Somaliland)	4.9 (3.2–7.4)	NA	4.5 (1.6–11.8)	NA
Sudan	10.2 (6.6–15.5)	NA	2.1 (1.4–3.2)	NA
Syria	19.1 (14.6–24.7)	42.6 (17.1–68.1)	5.9 (4.3–8.2)	NA
Tunisia	15.1 (12.3–18.4)	46.5 (43.9–49.2)	1.6 (0.8–3.1)	1.0 (0.6–1.5)
United Arab Emirates	12.1 (10.3–14.1)	25.5 (20.4–30.6)	3.6 (2.9–4.4)	1.6 (0.2–3.1)
West Bank§§	27.6 (21.3–35.1)	NA	8.7 (5.8–12.8)	NA
Yemen	4.2 (2.3–7.5)	NA	1.6 (0.8–3.1)	NA

* Source: 2008 MPOWER Report
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
NA–Data not available

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 6. (Continued)

WHO Region and Site	*Boy % (95% CI)*	*Male % (95% CI)*	*Girl % (95% CI)*	*Female % (95% CI)*
EUROPEAN REGION				
Bosnia and Herzegovina	14.3 (12.3–16.6)	49.3 (42.8–55.9)	9.4 (7.3–12.0)	35.1 (28.9–41.2)
Bulgaria	24.4 (20.2–29.2)	47.5 (39.2–55.7)	31.6 (25.9–37.9)	27.8 (19.7–35.9)
Croatia	21.7 (17.9–26.0)	38.9 (37.1–40.7)	25.6 (20.6–31.2)	29.1 (28.0–30.3)
Czech Republic	29.8 (25.1–35.0)	36.6 (30.0–43.2)	32.7 (27.6–38.1)	25.4 (17.9–32.8)
Estonia	28.2 (23.5–33.3)	49.9 (47.2–52.6)	26.2 (21.6–31.4)	27.5 (25.4–29.7)
Georgia	15.2 (9.9–22.8)	57.1 (48.4–65.8)	2.8 (1.0–7.8)	6.3 (2.4–10.1)
Hungary	21.5 (16.6–27.4)	45.7 (38.4–53.1)	23.6 (19.4–28.3)	33.9 (24.5–43.2)
Kyrgyzstan	6.8 (5.0–9.4)	46.9 (38.4–55.3)	2.2 (1.4–3.6)	2.2 (1.4–3.1)
Latvia	36.3 (30.9–42.1)	54.4 (45.6–63.1)	30.2 (24.1–37.0)	24.1 (20.9–27.3)
Lithuania	33.8 (29.4–38.6)	45.1 (37.9–52.3)	25.9 (21.2–31.2)	20.8 (18.0–23.7)
Montenegro	5.7 (4.3–7.6)	NA	4.4 (3.1–6.1)	NA
Poland (Urban)	22.1 (14.9–31.6)	43.9 (35.2–52.6)	20.1 (16.0–24.9)	27.2 (18.2–36.3)
Republic of Moldova	18.5 (15.0–22.6)	45.8 (38.4–53.1)	5.6 (4.3–7.2)	5.8 (4.2–7.5)
Russian Federation (Moscow)	23.3 (17.2–30.9)	70.1 (59.1–81.2)	27.2 (20.5–35.2)	26.5 (19.1–33.9)
Serbia	9.3 (6.3–13.4)	43.8 (39.1–48.6)	8.9 (6.6–11.9)	43.8 (39.1–48.6)
Slovakia	26.5 (23.2–29.9)	41.6 (34.6–48.6)	23.4 (20.7–26.4)	20.1 (14.4–25.9)
Slovenia	15.2 (10.7–21.2)	31.8 (25.7–37.9)	23.0 (18.7–27.9)	21.1 (16.2–26.1)
Ukraine (Kiev)	29.9 (22.5–38.4)	63.8 (53.6–74.0)	28.0 (20.3–37.2)	22.7 (16.5–28.9)

* Source: 2008 MPOWER Report
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
NA–Data not available

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 6 Gender difference in current cigarette smoking for adult males and females and adolescent boys and girls, by World Health Organization (WHO) region, WHO member state, geographic region, territory, special administrative region, or commonwealth and sex–MPOWER data for adults, GYTS data for youth

WHO Region and Site	*Boy % (95% CI)*	*Male % (95% CI)*	*Girl % (95% CI)*	*Female % (95% CI)*
REGION OF THE AMERICAS				
Antigua and Barbuda	2.7 (1.7–4.3)	NA	4.4 (2.3–8.2)	NA
Argentina (Capital Federal)	15.1 (11.2–20.2)	34.3 (30.8–37.8)	21.4 (15.2–29.3)	23.5 (20.7–26.3)
Bahamas	6.2 (3.8–10.1)	NA	3.7 (2.1–6.6)	NA
Barbados	14.3 (10.4–19.3)	16.9 (9.1–24.7)	9.3 (6.4–13.2)	2.3 (0.8–3.7)
Belize	11.7 (8.3–16.2)	NA	4.4 (2.6–7.5)	NA
Bolivia (La Paz)	20.3 (16.5–24.7)	33.8 (26.3–41.4)	12.0 (9.3–15.3)	26.1 (23.0–29.3)
Brazil (Curitiba)	10.0 (7.9–12.7)	NA	16.6 (13.9–19.8)	NA
Chile (Santiago)	28.0 (24.3–32.0)	41.7 (33.3–50.1)	39.9 (36.0–43.9)	30.5 (25.6–35.4)
Colombia (Bogota)	25.4 (21.0–30.3)	NA	26.6 (20.9–33.1)	NA
Costa Rica	9.4 (7.2–12.0)	26.1 (22.0–30.3)	9.7 (7.8–12.1)	7.3 (5.7–8.8)
Cuba (Havana)	11.2 (8.3–15.1)	36.1 (21.9–50.4)	8.8 (6.5–11.9)	26.4 (20.1–37.7)
Dominica	11.8 (8.1–16.9)	NA	9.6 (7.0–13.0)	NA
Ecuador (Quito)	23.2 (19.4–27.6)	23.6 (20.5–26.7)	18.1 (11.1–28.0)	5.6 (4.5–6.8)
Grenada	10.9 (7.4–15.8)	NA	9.5 (7.4–12.2)	NA
Guatemala (Guatemala City)	16.6 (12.6–21.5)	24.5 (20.5–28.4)	12.8 (10.1–16.2)	4.1 (3.2–5.1)
Guyana	11.0 (7.4–16.0)	NA	5.4 (3.1–9.3)	NA
Haiti (Port-au-Prince)	17.2 (12.4–23.5)	NA	17.7 (13.3–23.0)	NA

* Source: 2008 MPOWER Report
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
NA–Data not available

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 6. (Continued)

WHO Region and Site	*Boy % (95% CI)*	*Male % (95% CI)*	*Girl % (95% CI)*	*Female % (95% CI)*
Jamaica	20.6 (14.1–29.3)	18.8 (10.6–27.0)	10.9 (6.5–17.7)	7.6 (5.1–10.1)
Mexico (Mexico City)	26.3 (22.0–31.0)	36.9 (29.6–44.1)	27.1 (23.7–30.8)	12.4 (8.9–15.9)
Panama	5.9 (4.0–8.5)	NA	2.8 (1.7–4.6)	NA
Paraguay (Asuncion)	11.7 (9.3–14.5)	32.6 (28.8–36.4)	9.3 (6.6–12.8)	13.9 (11.8–16.0)
Peru (Lima)	16.7 (12.8–21.6)	NA	15.2 (11.0–20.7)	NA
Saint Lucia	17.0 (12.2–23.1)	25.4 (14.7–36.1)	9.6 (7.4–12.4)	9.2 (6.0–12.4)
Saint Vincent and The Grenadines	14.8 (9.8–21.7)	NA	9.5 (6.6–13.4)	NA
Suriname	9.3 (6.3–13.5)	NA	4.7 (2.7–8.2)	NA
Trinidad and Tobago	14.7 (10.9–19.6)	32.2 (19.3–45.1)	10.3 (6.9–15.1)	5.7 (3.8–7.7)
U.S. Virgin Islands††	3.0 (2.0–4.6)	NA	3.6 (2.4–5.5)	NA
United States of America	12.1 (10.5–13.9)	26.3 (23.2–29.5)	13.9 (12.3–15.8)	21.5 (18.0–25.0)
Uruguay	16.4 (13.5–19.8)	37.1 (32.7–41.6)	22.9 (20.1–26.0)	28.0 (24.0–32.0)
Venezuela (Lara State)	10.3 (6.2–16.7)	32.5 (26.7–38.4)	7.9 (5.6–11.2)	27.0 (21.1–32.9)
SOUTH-EAST ASIA REGION				
Bangladesh (Dhaka)	4.6 (2.8–7.5)	42.8 (34.6–50.9)	1.1 (0.4–2.9)	0.9 (0.4–1.4)
Bhutan	18.3 (13.8–23.8)	NA	6.3 (4.1–9.6)	NA
India	5.4 (4.3–6.7)	27.6 (22.2–32.9)	1.6 (1.0–2.6)	1.0 (0.6–1.3)
Indonesia (Jakarta)	28.4 (23.2–34.3)	62.1 (54.5–69.6)	3.1 (1.5–6.4)	4.0 (3.5–4.4)
Maldives	6.6 (4.6–9.6)	39.7 (32.2–47.2)	0.9 (0.4–2.0)	8.9 (6.0–11.9)
Sri Lanka	1.6 (0.7–3.7)	24.5 (19.7–29.2)	0.9 (0.2–3.5)	0.4 (0.2–0.7)

* Source: 2008 MPOWER Report
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
NA–Data not available

(Continued)

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Table 6. (Continued)

WHO Region and Site	*Boy % (95% CI)*	*Male % (95% CI)*	*Girl % (95% CI)*	*Female % (95% CI)*
WESTERN PACIFIC REGION				
China (Tianjin)	5.6 (3.9–7.9)	59.5 (54.6–64.5)	0.6 (0.3–1.1)	3.7 (2.6–4.9)
Fiji	6.7 (3.8–11.6)	23.6 (18.7–28.5)	3.1 (1.6–6.0)	5.1 (3.8–6.3)
Lao PDR (Vientiane Province)	12.0 (6.6–20.8)	61.1 (53.5–68.6)	0.4 (0.1–1.6)	13.6 (12.5–14.7)
Macau ¶¶	11.0 (8.1–14.8)	NA	9.8 (7.0–13.6)	NA
Mongolia	11.0 (7.6–15.6)	45.8 (32.1–59.5)	3.3 (1.4–7.3)	6.5 (4.7–8.2)
Northern Mariana Islands**	26.6 (23.6–29.9)	NA	31.5 (28.2–34.9)	NA
Palau	31.0 (26.9–35.5)	38.1 (28.1–48.2)	22.6 (18.1–27.8)	9.7 (5.1–14.3)
Philippines	23.4 (19.7–27.7)	38.9 (34.1–43.8)	12.0 (9.4–15.1)	8.5 (7.7–9.3)
Viet Nam (Hanoi)	5.0 (2.8–8.9)	42.9 (37.1–48.6)	1.0 (0.5–1.9)	2.2 (1.5–2.9)

* Source: 2008 MPOWER Report
§§ Geographic region
†† Territory of the United States
¶¶ Special Administrative Region of China
** Commonwealth in political union with the United States
NA–Data not available

effort to attract use by women, young girls and boys. Also, waterpipe use is a form of social smoking in that the pipes are often shared among friends and family at home or in bars and cafes that provide waterpipes to customers. Shisha smoking carries less of a cultural stigma than does cigarette smoking, particularly among Arab women (17). In these cultures, women and young girls are more likely to use a waterpipe than use other forms of tobacco (17).

Gender differences

Comparing cigarette smoking among boys and girls across the 100 sites, we found no change over time in 72 sites: in 42 sites boys had higher rates than girls; in 4 sites girls had higher rates than boys; and in 26 sites there was no gender difference (Table5). For the 28 sites that changed over time: 13 changed from boys having higher smoking rates than girls at Time 1 to no gender difference at Time 2; 11 changed from no gender difference at Time 1 to boys having higher smoking rates than girls at Time 2; and 4 changed from no gender difference at Time 1 to girls have higher rates than boys at Time 2.

For other tobacco use, we found no change over time in 62 of the 97 sites with data (Table 5). In 34 of the 62 sites, boys had higher use than girls and in the other 28 sites there was no gender difference. For the 35 sites that did change over time, 24 changed from boys having higher rates than girls at Time 1 to no gender difference at Time 2; and 11 changed from no gender difference at Time 1 to boys having a higher rate than girls at Time 2.

The findings from our report lend some support to the suggestion that smoking behavior among adolescent girls is increasing (18). We can compare results from the GYTS to adult prevalence data for 69 of the 100 sites (Table 6) (2). These data show that adult males have significantly higher rates than

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

adult females in 59 of the 69 sites. However, the GYTS data show that in 25 of the 69 sites there was no gender difference between boys and girls and in 60 of the 69 sites there was either no difference between the girl and adult female rate or girls had significantly higher rates than adult females. Target marketing and visibility of women smoking may have contributed to a change in cultural traditions and social influences, making smoking among women and young girls more acceptable. The tobacco industry targets women through advertisements showing smoking associated with independence, stylishness, weight control, sophistication, and power (19). In addition, sex-neutral brands such as Marlboro are marketed to women using independent and fun-loving imagery.

Conclusion

Results from this report show many challenges confront the global tobacco control effort. First, the majority of sites reported no change over time in prevalence of cigarette smoking (in 61 of the 100 sites) and use of other tobacco products (in 50 of the 97 sites with data). As the MPOWER Report concludes, 'Member States have a long way to go before they are effectively protecting their citizens from the tobacco epidemic' (2). Second, other tobacco use increased in 34 sites. In some countries and regions, this appears to be due to the recent fad among adolescents and young adults in smoking tobacco in waterpipes. Third, evidence was found supporting the idea that tobacco use among adolescent girls is likely increasing. This finding suggests tobacco control programs need to develop special programs directed toward tobacco use prevention among girls. Fourth, the Philippines results showed that political will can have a very strong impact on tobacco control. In the Philippines, the initial positive decrease in cigarette smoking from 2000 to 2004 was reversed between 2004 and 2007 due in part to a change in government and de-emphasis on tobacco control.

In the MPOWER Report, WHO acknowledged that few countries had adopted comprehensive tobacco control policies. However, Uruguay, Panama, Jordan and Sri Lanka are examples of the impact the adoption of effective, comprehensive tobacco control policies can have on reducing adolescent tobacco use. Hopefully, the findings from these countries will show other countries that by adopting comprehensive legislation, positive results can be achieved.

Notes

i. The school response rate is defined as: Number of participating schools/Number of selected schools. The class response rate is defined as: Number of participating classes/Number of selected classes. The student response rate is defined as: Number of students who completed the survey/Number of students enrolled in the class. The overall response rate is defined as: School response rate (times) Class response rate (times) Student response rate.

References

1. Jha P, Chaloupka FJ. Tobacco control in developing countries. Oxford: Oxford University Press; 2000.
2. World Health Organization. WHO Report on the Global Tobacco Epidemic, 2008 – The MPOWER package. Geneva: World Health Organization; 2008.
3. World Health Organization. WHO Framework Convention on Tobacco Control. Geneva: World Health Organization; 2003.
4. Centers for Disease Control and Prevention. Global Youth Tobacco Surveillance, 2000–2007. Surveillance Summaries, January 25, 2008. MMWR 2008; 57 (No. SS-1).
5. Shah BV, Barnwell BG, Bieler GS. Software for the statistical analysis of correlated data (SUDAAN): User's Manual. Release 7.5. 1997 (software documentation). Research Triangle Park, NC: Research Triangle Institute; 1997.
6. Hinkle DE, Wiersma W, Jurs SG. Applied statistics for the behavioral sciences. 5th edn. Boston, MA: Houghton Mifflin Co.; 2003.
7. Donner A and Klar N. Design and analysis of cluster randomization trials in health research. New York, NY: Qxford University Press Inc.; 2000.
8. Chaloupka F, et al. The taxation on tobacco products. In: Jha P, Chaloupka F, eds. Tobacco control in developing countries. Oxford, Oxford University Press; 2000: 237–72.
9. Belize Ministry of Health. National Tobacco Control Plan, 2007 (personal communication: Emmanuel Englebert and Lorraine Thompson, 12 February 2009).
10. Uruguay Ministry of Public Health. National Program for Tobacco Control, Law 18.256 (personal communication: Gabriela Olivera, 12 February 2009).
11. Slovenia Ministry of Health. Restriction on the Use of tobacco Products Act, enacted August 2007. Information from Slovenia Coalition for Tobacco Control and personal communication from Mojca Juricic.
12. Law No. 13. Republic of Panama. 13 January 2008. Digital Official Gazette No. 25966; 25 January 2008. Panama [Internet]. http://www.smoke-free.ca/warnings/laws/panamaLey%20N%C2%B0%2013%20de%2025%20de%20enero%20de%202008.pdf.

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

13. Milton MH, Maule CO, Yee SL, Backinger C, Malarcher AM, Husten CG. Youth Tobacco Cessation: A Guide for Making Informed Decisions. Atlanta: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention; 2004.
14. Jordan Ministry of Health. Monitoring Juveniles Behavior Naw No. 37, 2003 (personal communication: Malek Habashneh, 9 February 2009).
15. Miguel-Baquilod M, Fishburn B, Santos J, Jones NR, Warren CW. Tobacco use among 13–15 year olds in the Philippines, 2000–2003. CDC MMWR; 54(4): 94–7.
16. Prignot JJ, Sasco AJ, Poulet E, et al. Alternative forms of tobacco use. Int J Tuberc Lung Dis. 2008; 12(7): 718–27.
17. Soweid, RA. Lebanon: Waterpipe Line to Youth. Tobacco Control. 2005; 14: 363–4.
18. Mackay J, Eriksen M, Shafey O. The Tobacco Atlas. Atlanta, GA: American Cancer Society; 2006.
19. Waldron I, Bratelli G, Carriker L, Sung WC, Vogeli C, Waldman E. Gender differences in tobacco use in Africs, Asia, the Pacific and Latin America. Soc Sci Med. 1988; 27: 1269–75.

Appendix A: GYTS Support

Canadian Public Health Association: James Chauvin.

CDC/Office on Smoking and Health: Charles W. Warren, Juliette Lee, Veronica Lea, Ann Goding, Brandon O'Hara, Michelle Carlberg, Nathan Jones, Samira Asma, Matthew McKenna, Lorna English.

Research Triangle Institute Support: Mathew Bensen.

WHO/Tobacco Free Initiative

Headquarters: Douglas W. Bettcher, Edouard T d'Espaignet, Daniel Ferrante, Armando Peruga

WHO Regional Offices: African Region, Jean-Pierre Baptiste and Nivo Ramanandraibe; Eastern Mediterranean Region, Fatimah el Awa and Heba Fouad; European Region, Agis Tsouros and Kristina Mauer-Stender; Region of the Americas, Adriana Blanco and Renee Franklin-Peroune; South-East Asia Region, Khalil Rahman and Dhirendra N. Sinha; Western Pacific Region, Susan Mercado and Ali Akbar.

African Region: Larda Theodore Kangoye, Mathias Some, Drabo Maxime, Vitalien Kpdoa, Arouna Ouedraogo (Burkina Faso); Edith Wellington, S.O. Sackey (Ghana); Joyce Nato, Ezra Ouma Ogwell (Kenya); Patrick Kanyimbo, Adamson Muula, John Kapito, Esther Chaswa (Malawi); Elhadj Diop (Mauritania);Augusto Nunes (Mozambique); Daga Magagi (Niger); Cheikh Ibrahima Niang, Samba Cor Sarr (Senegal); Bharathi Viswanathan, Pascal Bovet (Seychelles); Dehran Swart, Fred Koopman (South Africa); Africa Magongo, David Pritchard, Sydney Nkambule (Swaziland); Osseni Tidjani, Abdou Gbadamassi (Togo); Lillian Mpabulungi, Frederick Musoke (Uganda); Richard Zulu, Mbiko Msoni (Zambia); Pepukai Chikukwa, Christopher Zishiri, Edwin G.V. Sithole, Shungu Munyati (Zimbabwe).

Region of the Americas: Colin O'Keiffe, Joan A. Moses (Antigua & Barbuda); Hugo A. Miguez (Argentina); Judith Scavella, Larrie Williams (Bahamas); Elaine M. King, Peter Wickham, Sean Daniel, Ruchelle Roach (Barbados); Kimani Avila (Belize); Franklin Alcaraz del Castillo (Bolivia); Luisa Goldfarb, Valeska Caralho Figueiredo, Adelemara Mattoso Allonzi, Leticia Casado Costa, Liz Maria de Almeida (Brazil); Luis Caris, Claudia Gonzalez Wedmaier (Chile); Carlos Jose Castro, Carolina Weisner, Constanza Parda Ramos, Raul Murillo (Colombia); Julio Bejarano (Costa Rica); Lucia Lances Cotilla (Cuba); Joan Henry (Dominica); Silvia Corella Ramirez, Patricio Jacome (Ecuador); Alister Antoine (Grenada); Mirna Gracia, Irma Perez, Miguel Garces, Ana Garces (Guatemala); Shradhanand Haviprashad (Guyana); Gilbert Jean Charles, Gerald Lerebours (Haiti); Karen Prendergast, Ellen Grizzle (Jamaica); Raydel Valdes Salgado, Blanca Villa-Contreras, Jesus Felipe Gonzalez Roldan, Luz Myriam Reynales Shigematsu (Mexico); Reina Roa (Panama); Graciela Gamarra (Paraguay); Alfonso Zavaleta (Peru); Elvina Lawrence, Cyprian Lande (Saint Lucia); Patsy Wyllie (Saint Vincent and the Grenadines); Gerold Vliet, Kris Rambali, M. Mangroo (Suriname); Leo Alleyne, Nicole Cooper (Trinidad & Tobago); Julia Sheen-Aaron (U.S.Virgin Islands); Raquel Magri, Gabriela Olivera (Uruguay); Ricardo Granero, Ligia de Pinto (Venezuela).

Eastern Mediterranean Region: Nevein Moneir Dous, Samy Ghanem, Mohamed Mehrez Mostafa (Egypt); Samah Eriqat, Salah Shaker Isa Soubani, Moein Al Kariry (Gaza Strip and West Bank); Ahmed Ali Bahaj, Ali Asghar Farshad, Hassan Azaripour Masooleh (Iran); Heba Ayoub, Imam Al Jaghbeer (Jordan); Sami Al Nasser (Kuwait); Georges Saade, Nagib Ghosn (Lebanon); Mohamed Salah, Ahmed Buni, Mohamed Salah (Libya); Sanaa

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016

Cherqaoui, Noureddine Chaouki, Abdelkhalek Moujarrade (Morocco); Sahar Abdou Helmy, Issa Al-Shuaili, Jawad Al-Lawati, Salah Al Muzahmi (Oman); Ahmad Moh'd Almulla, Ahmed Al-Ibrahim, Hamda Qotba (Qatar); Ali Sheikh Omar, Muse Yusuf Shirwae, Ali Sheikh Omaer Kabil, Mohamed Said Mohamed (Somalia); Ibrahim Abdelmageed Mohamed Ginawi, Ilham Abdalla Bashir (Sudan); Bassam Abu Al Zahab (Syria); Mohamed Nabil Ben Sahem, Alya Mahjoub Zarrouk, Mohamed Mokdad, Mongi Hamrouni (Tunisia); Bassam Abi Saab, Hala Haidar, Ayesha Al-Mutawa (United Arab Emirates); Ahmed Ali-Bahaj, Mohammed Al Khawlani (Yemen).

European Region: Tanja Coric, Dijana Mayer (Croatia); Hana Sovinova (Czech Republic); Kadi Lepp, Krystiine Liiv (Estonia); Iveta Pudule (Latvia); Antanas Gostautas (Lithuania); Krzysztof Przewoznjak (Poland); Elena Skvortsova, Konstantine Vitalievich Vyshinsky, Irina Parfenova, Andrei Demine (Russian Federation); Tibor Baska (Slovakia); Mojca Juricic (Slovenia); Konstantin Krasovsky, Nadezhda Polka, Tatiana Andreeva (Ukraine).

South-East Asia Region: Sonam Phuntsho (Bhutan); Dhirendra Sinha, Prakash Gupta, Urmi Sen, Surendra Shastri, Vendhan Gajalakshmi, Monika Arora, Mira Aghi, Rameshwar Sharma, M Prakasamma, G Gururaj, Arun Chaturvedi, Sanjeev Misra, Mihir Shah, Rajesh Dixit, R. Thulasidasan, and SK Jindal (India); Tjandra Yoga Aditama (Indonesia); Ahmed Waheed, Maimoona Aboobakar (Maldives); M R Pandey (Nepal); P W Gunasekera (Sri Lanka).

Western Pacific Region: Jiangping Sun, Jiang Yuan (China); Ilisapeci Movono, Mosese Salusalu (Fiji); Khattanaphone Phandouangsy, Anothay Kongsayasak, Ketkeo Boupha, Khampone Insixiengmay (Laos); Chan Tan Mui (Macao SAR); Isamu Abraham (Northern Mariana Islands); Marina Miguel-Baquilod (Philippines); Phan Thi Hai (Viet Nam).

Downloaded from ped.sagepub.com at SAGE Publications on June 21, 2016